

Discovering Safety Violations of Decision-Making in Autonomous Driving Systems from Accident-Free Traffic Scenarios

HAOXIANG TIAN, Nanyang Technological University, Singapore, Singapore and University of Chinese Academy of Sciences, Beijing, China

YUAN ZHOU, Zhejiang Sci-Tech University, Hangzhou, China

AN GUO, The Hong Kong Polytechnic University, Hong Kong, China

GUOQUAN WU, Key Laboratory of System Software (Chinese Academy of Sciences), Beijing, China and University of the Chinese Academy of Sciences, Beijing, China

MINGFEI CHENG, Singapore Management University, Singapore, Singapore

WEI WANG and **JUN WEI**, Key Laboratory of System Software (Chinese Academy of Sciences), Beijing, China and University of the Chinese Academy of Sciences, Beijing, China

TIANWEI ZHANG, Nanyang Technological University, Singapore, Singapore

XINGSHUO HAN, College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China

This work was done when Haoxiang Tian was at Nanyang Technological University as a visiting student.

Guoquan Wu, Wei Wang and Jun Wei are also affiliated with Nanjing Institute of Software Technology, University of Chinese Academy of Sciences, Nanjing, China.

This work was supported by National Natural Science Foundation of China under Grants (62472412, U20A6003, and 62572441), Major Project of ISCAS (ISCAS-ZD-202302), and Project of ISCAS (ISCAS-JCMS-202402). This work was also supported by National Research Foundation, Singapore and DSO National Laboratories under its AI Singapore Programme (AISG Award No: AISG2-GC-2023-008), and Ministry of Education, Singapore under its Academic Research Fund Tier 2 (Proposal ID: T2EP20223-0043). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of National Research Foundation, Singapore or DSO National Laboratories or Ministry of Education, Singapore.

Authors' Contact Information: Haoxiang Tian, Nanyang Technological University, Singapore, Singapore and University of Chinese Academy of Sciences, Beijing, China; e-mail: haoxiang.tian@ntu.edu.sg; Yuan Zhou, Zhejiang Sci-Tech University, Hangzhou, China; e-mail: yuanzhou@zstu.edu.cn; An Guo, The Hong Kong Polytechnic University, Hong Kong, China; e-mail: an.guo@polyu.edu.hk; Guoquan Wu (corresponding author), Key Laboratory of System Software (Chinese Academy of Sciences), Beijing, China and University of the Chinese Academy of Sciences, Beijing, China; e-mail: gqw@otcaix.iscas.ac.cn; Mingfei Cheng, Singapore Management University, Singapore, Singapore; e-mail: snowbirds.mf@gmail.com; Wei Wang, Key Laboratory of System Software (Chinese Academy of Sciences), Beijing, China and University of the Chinese Academy of Sciences, Beijing, China; e-mail: wangwei@otcaix.iscas.ac.cn; Jun Wei, Key Laboratory of System Software (Chinese Academy of Sciences), Beijing, China and University of the Chinese Academy of Sciences, Beijing, China; e-mail: wj@otcaix.iscas.ac.cn; Tianwei Zhang, Nanyang Technological University, Singapore, Singapore; e-mail: tianwei.zhang@ntu.edu.sg; Xingshuo Han (corresponding author), College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China; e-mail: xingshuo.han@nuaa.edu.cn.



This work is licensed under Creative Commons Attribution International 4.0.

© 2026 Copyright held by the owner/author(s).

ACM 1557-7392/2026/9-ART289

<https://doi.org/10.1145/3787978>

Safety testing serves as the fundamental pillar for the development of Autonomous Driving Systems (ADSs), and decision-making plays a key role in ADSs. To ensure the safety of ADSs, it is paramount to generate a range of critical test scenarios to test the safety of decision-making in ADSs. While existing research primarily focuses on reproducing real-world traffic accidents in simulation environments to create test scenarios, it is essential to highlight that many of these accidents do not result in safety violations of decision-making in ADSs due to the differences between human driving and autonomous driving. More importantly, we observe that some accident-free real-world scenarios can lead to misbehaviors of ADSs. Therefore, orthogonally to existing work, it is equally important to discover safety violations of ADSs from routine traffic scenarios (i.e., accident-free scenarios) to ensure the safety of Autonomous Vehicles (AVs).

We introduce CRISER, a novel methodology to achieve the above goal. It automatically generates abstract and concrete scenarios from real-traffic videos where human-driving worked safely. Based on them, CRISER discovers safety violations of the ADS's decision-making in semantic equivalent scenarios (i.e., the test scenarios with the same semantics as the original accident-free traffic videos). Specifically, CRISER enhances the ability of Large Multimodal Models (LMMs) to accurately extract scenario semantics from accident-free traffic videos and generate test scenarios by multi-modal few-shot Chain-of-Thought (CoT). Based on them, CRISER explores the behavior differences between the ego vehicle (i.e., the vehicle connected to the ADS under test) and human-driving in semantic equivalent scenarios. During the exploration search, CRISER keeps the semantic consistency of test scenarios with accident-free traffic videos and explores the universality of discovered safety violations of the ADS. We implement and evaluate CRISER on the industrial-grade Level-4 ADS, Apollo. The experimental results demonstrate that CRISER can accurately extract scenario semantics and generate test scenarios from traffic videos, and effectively discover distinct types of safety violations of Apollo's decision-making in accident-free traffic scenarios.

CCS Concepts: • **Software and its engineering** → **Software verification and validation**;

Additional Key Words and Phrases: Autonomous Driving System Test, Accident-free Traffic Scenario

ACM Reference format:

Haixiang Tian, Yuan Zhou, An Guo, Guoquan Wu, Mingfei Cheng, Wei Wang, Jun Wei, Tianwei Zhang, and Xingshuo Han. 2026. Discovering Safety Violations of Decision-Making in Autonomous Driving Systems from Accident-Free Traffic Scenarios. *ACM Trans. Softw. Eng. Methodol.* 35, 9, Article 289 (September 2026), 44 pages.

<https://doi.org/10.1145/3787978>

1 Introduction

Autonomous Driving System (ADS) testing is one of the most important tasks for the safety assurance of **Autonomous Vehicle (AV)** software [26]. As shown in Figure 1, the typical ADS pipeline comprises four key components: (1) *Sensing* integrates multi-modal sensors (e.g., camera, LiDAR) to capture real-time environment; (2) *Perception* processes sensor inputs through networks to detect and classify objects such as vehicles, pedestrians, and traffic signals; (3) *Decision-Making* synthesizes perception outputs with map data to generate navigational trajectories while adhering to safety constraints; (4) *Control* converts high-level decisions into throttle, brake, and steering actuations.

Among these components, decision-making constitutes the critical reasoning layer of ADSs, responsible for safe navigation through two tightly coupled submodules: trajectory prediction and motion planning. Trajectory prediction analyzes perceived objects' dynamics to forecast their future movements, while motion planning synthesizes this information with road geometry and traffic rules to compute AV's optimal path. This bidirectional interaction between these two submodules forms a continuous loop where they feed back to each other to complete the decision-making

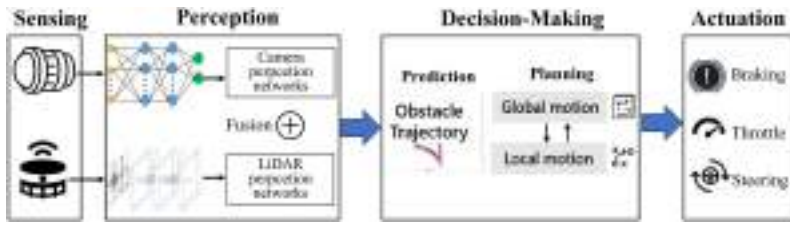


Fig. 1. The pipeline of a production-level ADS (L-4).

process. The control operations and actions of AVs are sensitive to the performance of the decision-making, as any mistake it makes will directly lead to AV's undesirable driving behaviors.

It is essential to perform thorough testing on the decision-making of the ADS before deploying it in the real world [21]. However, physical testing on public roads takes huge costs and increases the risk of accidents. Alternatively, simulation testing can create massive scenarios with extremely low costs. Several companies [1, 7] and academic researchers [29, 32, 56] have reproduced real-world accidents in the simulator to construct test scenarios to discover the safety violations of the decision-making in ADSs. In their simulation testing, the testers assign the ego vehicle controlled by the ADS under test following the route of a vehicle involved in the collision and define the trajectories of other participants as those of other vehicles and pedestrians in the accident.

However, this strategy has two shortcomings due to the fundamental differences between the design logic of decision-making of ADSs and human drivers [20, 28, 31]. First, in real-traffic accidents, more than 55% crashes [9] are not effectively linked to the safety violations of decision-making of ADSs, such as those caused by alcohol, fatigue, drivers' physical and internal illness [19, 49], which will not occur to the decision-making of ADSs. Second, due to the inherent differences in decision-making between human drivers and ADSs, some safe human-driving traffic scenarios can actually cause safety violations of the decision-making of ADSs.

Figure 2 shows one example. In the human-driving case, vehicle *S* is driving at a very slow speed. The human driver of the following vehicle *H* makes the decision to overtake *S* and no accident occurs. We apply this scenario to test the industrial ADS, Apollo [8], where NPC vehicle *N* is moving slowly followed by the ego vehicle *E*. The decision-making of *E* regards *N* as stationary due to its very slow speed, and generates the future path for overtaking *N*. However, at runtime, the motion of *N* makes its distance to *E* vary slightly, which misleads the decision-making of *E* to change the path into not changing, causing *E* to get stuck for a long time and finally collide with *N*. Consequently, *orthogonally to existing works that test ADSs in traffic accident scenarios, it is equally important to conduct safety evaluations of ADS's decision-making in accident-free human-driving scenarios, which is rarely investigated.*

Motivated by the above observations, this article proposes CRISER, a novel approach that discovers safety violations of decision-making of ADSs from real accident-free human-driving traffic scenarios. Our source of traffic scenarios is the automotive videos, which are commonly available and completely record the static and dynamic surrounding objectives [30]. The goal of CRISER is to discover safety violations of decision-making in the ADS from traffic situations where human drivers drive safely. In this article, *semantic equivalent scenarios* are the test scenarios that preserve the same scenario semantics as the original accident-free human-driving traffic scenario. CRISER targets the intermediate level of driving-scenario *semantics*, and focuses on leveraging dynamic participants (e.g., NPC vehicles, pedestrians) to test the decision-making of ADS. The formal definitions of the scenario semantics will be given in Section 2.

There exist some challenges for CRISER to solve the above tasks:

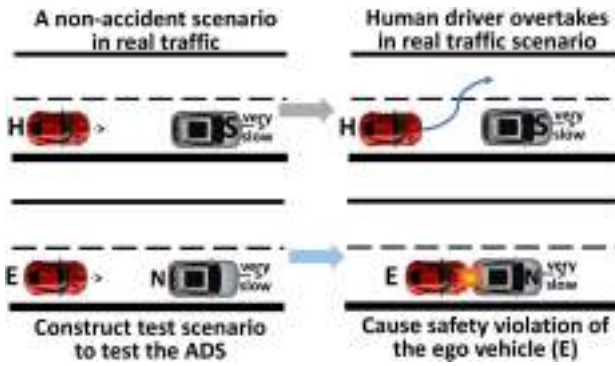


Fig. 2. An example of a real-traffic scenario without accident that can cause safety violations of ADSs.

- *Challenge 1:* How to design and implement a lightweight method to understand and extract scenario semantics of traffic videos, and generate scenarios to test the decision-making of ADS?
- *Challenge 2:* How to search for critical semantic equivalent scenarios to find safety violations of ADS decision-making? Since human-driving is known to make safe decisions and drive safely within the scenario semantic context of accident-free traffic scenarios, the critical semantic equivalent scenarios in which ADS behaves significantly differently from human-driving thus reveal safety violations of the ADS decision-making in the scenario semantic contexts where human-driving remains safe.

CRISER consists of two novel techniques to address the above challenges. *For challenge 1*, we propose a method for *traffic video-based scenario understanding and generation*. Inspired by the great contextual text-image understanding and reasoning capability of **Large Multimodal Models (LMMs)** [15, 37, 44], CRISER first utilizes an LMM to extract scenario semantics of traffic videos. As current LMMs do not demonstrate great performance in understanding traffic videos [38, 43, 55], CRISER utilizes optical flow change analysis to enhance its ability in interpreting and extracting scenario semantics. Then CRISER utilizes the multi-modal few-shot **Chain-of-Thought (CoT)** to guide the LMM to generate test scenarios according to the extracted scenario semantics.

For challenge 2, we need to address three main difficulties: (1) effectively searching for critical semantic equivalent scenarios, (2) preserving semantic equivalence during the search, and (3) revealing generalizable ADS decision-making safety violations in semantic equivalent scenarios. To this end, we propose a *specification-based dual-layer exploration framework for safety-critical semantic equivalent scenarios*. Specifically:

- To effectively search for critical semantic equivalent scenarios, the outer-layer exploration of CRISER searches for test scenarios that maximize the behavioral differences between human-driving in accident-free traffic scenarios and the ego vehicle in semantic equivalent scenarios. This enables the discovery of ADS decision-making safety violations in situations where the human-driving remains safe. To ensure that the detected safety violations originate from the ADS's decision-making rather than its perception module, CRISER directly feeds the ground-truth data of the test scenarios into the ADS's decision-making module.
- To reveal generalizable ADS decision-making safety violations in accident-free semantic equivalent scenarios, for the discovered **Safety Violation Scenario (SV)**, the inner-layer exploration of CRISER further examines variations in the trajectories of dynamic participants to generate several concrete scenarios of the accident-free scenario semantics. Then CRISER

tests and assesses whether the discovered ADS's safety violation persists across them, rather than occasional occurrences in specific concrete scenarios of this scenario semantics.

- To preserve semantic equivalence during the dual-layer exploration process, CRISER employs formal semantic specifications to determine whether the semantics of generated test scenarios are equivalent to the semantics of the original accident-free human-driving scenarios.

To evaluate the performance of CRISER, we conduct evaluations on the industrial L-4 ADS, Baidu Apollo [8]. Experimental results show that CRISER can correctly extract scenario semantics and generate semantic equivalent scenarios from real-traffic videos. Furthermore, it can effectively generate safety-critical semantic equivalent scenarios and find 10 distinct types of safety violations in Apollo's decision-making from accident-free human-driving scenarios, 7 of which are never discovered by existing **State-of-the-Art (SOTA)** approaches. To further validate the generalizability of our discovered decision-making safety violations of ADS, we use the critical semantic equivalent scenarios from Apollo to test another industrial ADS, Autoware [18]. The results show that 10 types of ADS decision-making safety violations discovered by CRISER are general in different ADSs. Moreover, to verify that the identified safety violations are independent of the simulator's characteristics, we reproduce the discovered critical semantic equivalent scenarios in another simulator, Carla [24], to test the decision-making of Apollo and Autoware. The results show that the discovered SVs are consistent across different simulators.

The contributions of this article are as follows:

- We introduce CRISER, a novel approach to automatically find safety violations of decision-making in ADSs from accident-free traffic videos.
- CRISER enhances LMM's ability to accurately extract scenario semantics of traffic videos by optical flow analysis, and to generate test scenarios by multi-modal few-shot CoT.
- CRISER designs a specification-based dual-layer exploration to test and assess the decision-making differences between the ego vehicle in semantic equivalent scenarios and human-driving in accident-free traffic videos.
- We test CRISER on the industrial L4 ADSs. The experimental results demonstrate the effectiveness of CRISER. Compared with SOTA approaches, CRISER can extract scenario semantics and generate semantic equivalent scenarios from accident-free automotive videos more accurately, and discover distinct types of safety violations in the decision-making of different ADSs.

2 Preliminaries

2.1 Formal Definition of Scenario Semantics

The semantics of scenario (S_i) in CRISER are formally represented as:

$$S_i = \{R_i, E_i, P_i, C_i\}, \quad (1)$$

$$R_i = \{rt_i, ln_i, rd_i, ts_i\}, E_i = \{dt_i\}, P_i = \{p_1, p_2, \dots, p_m\}, p_m = pt_i, bh_i, rp_i, dd_i, \quad (2)$$

where S_i consists of: R_i represents the road topology in S_i ; E_i represents the ego vehicle in S_i ; P_i represents the dynamic participants in S_i ; and C_i contains the constraints among R_i , E_i , and P_i . Their attributes are given in Table 1.

2.2 Semantic Equivalence

The semantic equivalence between the accident-free human-driving scenario S_i and the generated test scenario S_j is defined as:

$$S_i \equiv S_j \iff \forall c \in (R_i \cup E_i \cup P_i), \phi_c(S_i) = \phi_c(S_j),$$

Table 1. The Formal Definitions of Scenario Semantics

Attribute	Attribute name	Domain of attribute
rt_i	Road type	$\mathbb{R} = \{\text{straightroad}, \text{turnaround}, \text{crossroad}, \text{T-junction}\}$
ln_i	Number of lanes	$\mathbb{L} = \{1, 2, 3, \dots, 8\}$
rd_i	Road direction	$\mathbb{D} = \{\text{directional}, \text{unidirectional}\}$
ts_i	Traffic sign	$\{\text{traffic light}, \text{stop sign}, \text{speed limit}\}$
dt_i	Driving task	$\mathbb{B} = \left\{ \begin{array}{l} \text{follow lane}, \text{change left}, \text{turn left}, \text{turn} \\ \text{right}, \text{cross}, \text{accelerate}, \text{decelerate}, \text{brake} \end{array} \right\}$
pt_i	Participant type	$\mathbb{P} = \{\text{NPC vehicle}, \text{pedestrian}\}$
bh_i	Participant behavior	$\mathbb{B} \cup \{\text{walk along}, \text{walk across}, \text{stand}\}$
rp_i	Initial position relative to ego vehicle	$\mathbb{I} = \left\{ \begin{array}{l} \text{front}, \text{left ahead}, \text{right ahead}, \text{left behind}, \\ \text{right behind}, \text{left vertical}, \text{right}, \text{vertical}, \\ \text{opposite}, \text{opposite left}, \text{opposite right} \end{array} \right\}$
dd_i	Behavior direction relative to ego vehicle	$\mathbb{D} = \{\text{ahead}, \text{left}, \text{right}, \text{opposite}\}$
C_i	Semantic constraints	$\mathbb{C} = (rt_i = \text{straight road}) \implies (dt_i \in \mathbb{B} - \{\text{cross}, \text{turn right}, \text{turn left}\}) \cap (rp_i \in \mathbb{I} - \{\text{left vertical}, \text{right vertical}, \text{opposite}, \text{opposite left}, \text{opposite right}\})$

where $\phi_c(S_i)$ denotes the value of the attribute c in S_i . We focus on semantic equivalence in initial conditions and participant behaviors during scenario execution, including road topology, ego vehicle's driving task, initial positions of participants relative to ego vehicle, and types of participant behaviors. These elements define the foundational context of the test scenario. Semantic equivalence of initial conditions and participant behaviors during scenario execution between test scenarios with accident-free traffic scenarios ensures that the ADS is tested under a similar situational setup as the human-driving.

3 Background and Related Work

3.1 Semantics of Test Scenarios

The semantics of ADS test scenarios refer to the formal definition and interpretation of scenario elements that describe driving situations [33, 57], including factors such as the road, traffic participants, and environmental conditions. As our focus is on the decision-making of the ADS, the semantics of the test scenarios in this work contain high-level abstraction of traffic elements influencing decision-making, including dynamic participants (e.g., vehicles, pedestrians) and static road elements (e.g., lanes, traffic signs).

3.2 LMMs in Understanding Traffic Videos

LMMs have demonstrated great capability of understanding traffic scenarios from contextual text-image input [11, 15]. However, they have challenges in understanding traffic videos due to the dynamic nature of events, which require analyzing continuous changes in both visual and spatial information over time. Some existing research [38] emphasizes the challenges faced by LMMs in effectively transferring their textual capabilities to real-world video dynamics, mainly due to the computational challenges and limitations in capturing fine-grained temporal details [43, 55].

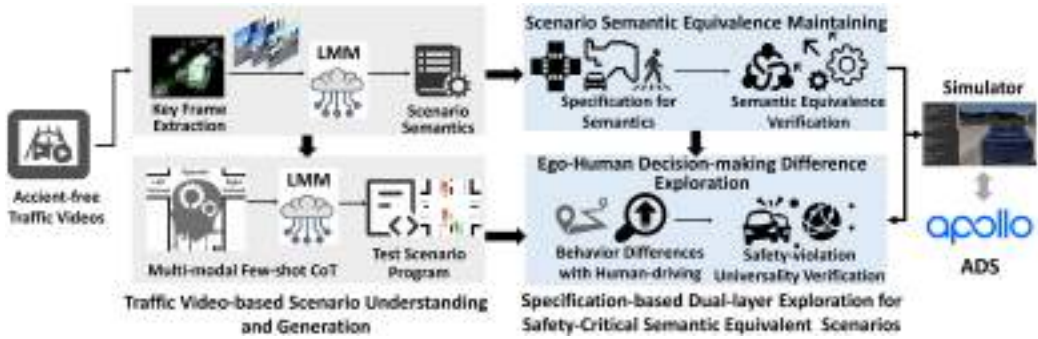


Fig. 3. The overview of CRISER.

3.3 Test Scenario-Based ADS Testing

The effectiveness of ADS testing highly depends on the quality of generated scenarios. How to effectively generate safety-critical scenarios has received significant attention in recent years.

Some works reproduce traffic accidents from crash databases as the test scenarios [29, 32, 51, 56]. Specifically, AC3R [29] uses domain-specific ontology and natural language processing to extract information from police crash reports and reconstruct corner cases of crash accidents. SoVAR [32] and LeGEND [51] utilize the Large Language Model to extract comprehensive information from accident reports and generate test scenarios based on them. CRISCO [53] mines behavior patterns from collision and collision-related trajectories, and generates test scenarios by constraint solving. However, both methods require crash reports that follow rigid narrative standards and record detailed text information of accidents, not applicable to free-form scenarios, e.g., automotive videos from real-life vehicles, to executable concrete scenarios. Zhang and Cai [56] train a panoptic segmentation model, M-CPS, to extract effective information from the accident videos and recover traffic participants in the simulator. However, it is time-consuming to train and optimize the extraction model. Furthermore, the accuracy of M-CPS in key information extraction is not high. More importantly, a common limitation of these methods is that they are only applicable for traffic accidents, but not feasible to generate safety-critical scenarios from massive regular traffic situations. Traffic crashes are rare in real life and most of them cannot create effective challenges to ADSs [19, 49]. Different from and orthogonal to these works, CRISER generates test scenarios from traffic accident-free videos, explores the differences of ego vehicle's decision-making and human-driving in the accident-free traffic situations, and discovers the safety violations of the decision-making in the ADS.

4 Approach

Figure 3 shows the overview of CRISER, including two parts:

- (1) *Traffic video-based scenario understanding and generation* (Section 4.1): The input of CRISER is automotive videos. It first utilizes optical flow analysis and state interpolation to enhance the LMM's ability in understanding and extracting scenario semantics from accident-free traffic videos. Based on these, CRISER utilizes multi-modal few-shot CoT to guide LMMs to generate test scenarios conforming to the semantics of the input traffic video.
- (2) *Specification-based dual-layer exploration for safety-critical semantic equivalent scenarios* (Section 4.2): Based on the extracted scenario semantics and generated test scenarios, CRISER designs a specification-based dual-layer exploration to discover safety violations of the ADS in the semantic equivalent scenarios of accident-free traffic videos. Specifically, to

test the decision-making of the ADS, CRISER feeds the ground truth data of generated test scenarios into this module, and tests the driving behaviors of the ego vehicle during test scenario execution. The outer layer explores the behavior differences between the ego vehicle in semantic equivalent scenarios and human-driving in the original accident-free traffic videos. For each discovered safety violation of ego vehicle's decision-making, the inner layer verifies whether it's universal in the semantic equivalent scenarios. During the dual-layer exploration, to ensure the semantics of generated test scenarios are equivalent with accident-free traffic videos, CRISER describes specifications of scenario semantics, and uses them to verify whether the semantics of the newly generated test scenarios violate the semantics of the original traffic video.

4.1 Traffic Video-Based Scenario Understanding and Generation

Traffic videos provide rich and complete information of real-traffic scenarios, and have become one of the most common and accessible data sources of real-world traffic [30]. To generate test scenarios from real traffic, CRISER first understands semantics of traffic scenarios from automotive videos. Past works have demonstrated that LMMs, represented by GPT-4V [13, 14], have strong capabilities in understanding the traffic elements in images [11, 15]. However, they have not demonstrated convincing performance in understanding traffic videos [38, 43, 55]. To address this gap, CRISER designs a motion change-based key frame extraction technique, which uses optical flow analysis and state interpolation to identify motion changes of participants in the traffic video as key frames. Then it leverages the LMM to understand and extract the key information of the scenario from the key frame sequence, and formulates to constitute the scenario semantics of the traffic video.

To generate corresponding test scenarios of the extracted scenario semantics, CRISER utilizes multi-modal few-shot CoT to guide the LMM for the generation, by leveraging LMM's contextual text-image learning and reasoning capability [45].

4.1.1 Motion Change-Based Key Frame Extraction. Previous approaches to key frame extraction, such as uniform sampling [34] and attention-based methods [25, 48], achieve varying degrees of success in identifying important frames. However, as real-world traffic environments are complex with various participant behaviors, these methods struggle to balance between computational efficiency and the ability to capture non-uniform frame transitions.

CRISER identifies key frames based on motion changes of vehicles and pedestrians. Specifically, it leverages optical flow analysis to obtain the motion states of vehicles and pedestrians over time in traffic videos. For each frame f , based on the motion states of its former and next frames, CRISER employs linear interpolation to estimate the intermediate state between them. If the participant's motion state does not change between two frames, this intermediate state should be equal or close to the participant's actual motion state at the frame f . For the frames where the motion states of participants significantly change, they are sequenced as key frames of the traffic video.

First, CRISER segments the traffic video into the sequence of scenario frames. It leverages optical flow analysis [54] to separate the moving objects from the background and generate optical flow field vector for the moving objects. To reduce the computation cost and improve the accuracy of obtaining dynamic object information, CRISER converts the sequence of frames into the gray-scale format, and generates the optical flow field vector that contains information about the motion states of objects in consecutive frames. The optical flow field vector of a scenario is formed by the displacement of corresponding pixels between consecutive frames caused by motion objects. For the traffic video segmented as N frames, the optical field vector consists of N motion state vectors. At frame f , the motion state vector is represented as $\vec{S}_f$, where K represents the number of dynamic

objects at the frame, $\mathbb{X}_i$ and $\mathbb{Y}_i$ are the coordinates relative to the camera origin of the pixels in the i th dynamic object, and $\mathbb{V}\mathbb{X}_i$ and $\mathbb{V}\mathbb{Y}_i$ are the horizontal and vertical velocity components of the object. Here, CRISER adopts Lucas Kanade algorithm [41] in the optical flow analysis:

$$\vec{S}_f = \begin{bmatrix} \mathbb{X}_1^f & \mathbb{Y}_1^f & \mathbb{V}\mathbb{X}_1^f & \mathbb{V}\mathbb{Y}_1^f \\ \mathbb{X}_2^f & \mathbb{Y}_2^f & \mathbb{V}\mathbb{X}_2^f & \mathbb{V}\mathbb{Y}_2^f \\ \dots & \dots & \dots & \dots \\ \mathbb{X}_K^f & \mathbb{Y}_K^f & \mathbb{V}\mathbb{X}_K^f & \mathbb{V}\mathbb{Y}_K^f \end{bmatrix}.$$

Based on the optical flow vectors of the traffic scenario, CRISER uses linear interpolation [47] to identify the significant changes of motion states between frames. For the motion state vector $\vec{S}_f$ at frame f , CRISER selects the vectors of its former frame $\vec{S}_{f-1}$ and next frame $\vec{S}_{f+1}$. Then it uses linear interpolation to compute the intermediate vector between $\vec{S}_{f-1}$ and $\vec{S}_{f+1}$ in a linear progression. The intermediate state $\vec{P}'_f$ is computed as:

$$\vec{P}'_f = (1 - \alpha)\vec{S}_{f-1} + \alpha\vec{S}_{f+1}, \alpha \in [0, 1],$$

where α represents the interpolation parameter. CRISER measures the difference between the actual motion state vector $\vec{S}_f$ and the intermediate vector $\vec{P}'_f$ to identify the motion changes at frame f . The non-linear changes of dynamic objects correspond to key events (e.g., sudden stops, lane changes, accelerations) in their movements. Therefore, for all frames of the scenario, the non-linear changes are identified to extract key frames, which are added to the key frame sequence if the interpolated state deviates significantly from the real state. Formally, the difference between the actual motion state vector $\vec{S}_f$ and the intermediate vector $\vec{P}'_f$ is calculated as follows:

$$\Delta M(\vec{S}_f, \vec{P}'_f) = \sum_{x,y} \left\| L_{x,y}(\vec{S}_f, \vec{P}'_f), O_{x,y}(\vec{S}_f, \vec{P}'_f) \right\|,$$

where L_x and L_y are the differences of motion state vectors in the horizontal and vertical positions, respectively; O_x and O_y are the differences of motion state vectors in the horizontal and vertical velocities, respectively. $\|\cdot\|$ denotes a L2-norm to quantify the difference between motion states. A significant deviation (i.e., above a threshold τ) exists between the actual motion state vector and the expected linear motion state vector.

Data collected in real traffic by the automotive camera may contain some factors that interfere with the optical flow analysis, mainly including slow movers, occlusions, and camera motion. CRISER mitigates these limitations through a combination of flow normalization, object-level segmentation, and motion consistency checks, as detailed below.

- Slow movers: For each detected object, we normalize the per-frame flow vector by the mean background flow magnitude to avoid suppressing small but consistent motion. A minimum-motion threshold (e.g., a tiny proportion b_f of the background flow magnitude) is used to preserve slow-moving entities such as pedestrians. This adaptive threshold prevents “zeroing out” subtle motion that would otherwise be lost during optical flow quantization.
- Occlusions: When object-level masks from the segmentation branch temporarily disappear, the flow trajectory is linearly extrapolated for up to three consecutive frames based on the last two valid flow states. If the object reappears within this window and its new flow direction deviates by less than an angle threshold l_r , continuity is maintained. Otherwise, a new trajectory ID is assigned to avoid incorrect flow merging.
- Camera motion: CRISER compensates for the automotive camera motion by estimating a global homography between consecutive frames using **Speeded-Up Robust Features (SURF)**

keypoints extracted from static background regions. SURF [22] is a fast and rotation-invariant keypoint detector that identifies stable feature points suitable for frame-to-frame matching. By matching these keypoints, CRISER computes the global transformation of the camera and subtracts the corresponding global flow field from the dense optical-flow map before motion-state inference. This correction ensures that only residual motion relative to the background is analyzed, effectively removing background-induced flow and stabilizing object-level motion extraction.

4.1.2 Scenario Semantics Extraction and Construction. Based on the extracted key frames of a traffic video, CRISER utilizes the LMM to understand and extract the key information of the scenario. As mentioned in Section 3.1, CRISER extracts the information including road topology, driving task of ego vehicle, behaviors of NPC vehicles and pedestrians, and their relative positions to the ego vehicle. The designed linguistic pattern of the extraction prompt is:

You are an autonomous driving expert who specializes in identifying dynamic objects in traffic scenarios. I will show you a series of traffic pictures taken by the camera of vehicle you are driving. These pictures are from the same scenario. Please use concise and structured language to describe the following objects in the scenario: road topology, behaviors of the vehicle you are driving, behaviors and positions of traffic participants (all vehicles and pedestrians, other signals and obstacles within the visible range).

For the extracted results, CRISER organizes them in a hierarchical representation as the semantics of the traffic video. Table 2 shows the descriptions and examples of the scenario semantics for testing decision-making of ADSs.

4.1.3 Test Scenario Generation. Based on the scenario semantics, CRISER designs a lightweight method to guide the LMM to generate test scenarios by CoT. In the simulation testing of the ADS, test scenarios are normally described by *scenario programs*, which are executable in the simulator. To mitigate the hallucination problem in using the LMM, CRISER utilizes the multi-modal few shots to improve the LMM's ability to generate representations for the participants' behaviors with their relative positions (reference to the ego vehicle) specified in the extracted semantics, and uses specification-based check to validate the output of the LMM.

First, we study the practice of test scenario programs [3, 23, 27, 35] and conclude that the elements related to decision-making test in them can be decomposed into five stages: road statement, ego driving task statement, NPC vehicle statement, pedestrians statement, and assertion statement.

- *Stage 1: Road statement* specifies a road conforming to the road topology in scenario semantics.
- *Stage 2: Ego driving task statement* defines the type, initial position and destination of ego vehicle. The initial position and destination make ego vehicle plan a route that corresponds to *ego driving task* in scenario semantics.
- *Stage 3: NPC vehicle statement* defines the types and motions of NPC vehicles, making them perform the corresponding participant behaviors in scenario semantics.
- *Stage 4: Pedestrians statement* defines the motions of pedestrians, making them perform the corresponding participant behaviors in scenario semantics.
- *Stage 5: Assertion statement* is to verify whether the ADS behavior aligns with safety requirements.

The motion representation of the NPC vehicle and pedestrian in the scenario program can be two types: the sequence of waypoints and the behavior with its relative position.

Based on the above five stages, CRISER uses the multi-modal few-shot CoT to instruct the LMM to solve them, generating test scenario programs for the scenario semantics. This is inspired by our practice that LMMs perform well in identifying scenario semantics from motion changes in traffic

Table 2. An Example of Scenario Semantics

Attribute	Description	Example of value
Road topology		
Road type	The type of the road	straight
Lane num	The number of lanes on the road	4
Direction	The directions of the lanes	bidirectional
Traffic sign	The traffic sign on the road	light = "green"
Ego driving task		
Behaviour	Behavior of ego vehicle's driving	follow lane
Direction	Direction of ego vehicle's driving	forward
Initial conditions and behaviors of dynamic participants		
Type	Pedestrian, or vehicle model	suv
Behaviour	Behavior of the participant	decelerate
Relative position	Initial position relative to ego vehicle's initial position	ahead

Table 3. Prompts for Waypoint-Based Scenario Program Generation

Type	Sample of prompt patterns
(1)	Please generate n test scenario programs according to scenario semantics. <Scenario program example>, <Introduction of parameters of scenario program>
(2)	On the input road, S and D are the examples of ego vehicle's initial position and destination for "follow lane" task. Please generate them for the given ego driving task.
(3)	The road is divided into divisions relative to ego vehicle's initial position.
(4)	On the road, a trajectory of an NPC vehicle that performs "follow lane" ahead, is defined as: <example trajectory>. Please generate a trajectory for each participant behavior.

scenarios, indicating that it fully understands the general knowledge of scenario semantics. Considering the two types of motion representations, CRISER employs the CoT process to instruct the LMM in generating two forms of scenario description programs. The first form of scenario programs defines the relative positions and motions of the ego vehicle and dynamic participants through trajectory waypoints (we call them *waypoint-based scenario programs*, such as AVUnit [3]). Another form of scenario programs explicitly declares the motions of the ego vehicle and dynamic participants through keywords (we call them *keyword-based scenario programs*, such as TARGET [23]). The waypoint-based scenario program generation of CRISER requires a more sophisticated CoT reasoning process and a more rigorous hallucination check, but it facilitates the subsequent scenario exploration of CRISER. The CoT reasoning for keyword-based scenario program generation of CRISER is comparatively simplified and easier to validate, while the generated scenario programs cannot be evolved without altering their semantics.

The process of CRISER's CoT-guided waypoint-based scenario program generation is:

- (1) *Instruct LMM to learn about the task context*: The CoT starts by informing the LMM of the task that generates test scenario programs from the input scenario semantics. The prompt pattern is given in Table 3 (1).
- (2) *Select the road (for Stage 1)*: CRISER selects the road from map file according to the road topology in scenario semantics, and retrieves information of lanes in the road, including lane ID, lane direction and lane length.
- (3) *Guide LMM to assign ego vehicle's initial position and destination (for Stage 2)*: Providing the picture of selected road, the few-shot for this step is generated by an example for the basic task *follow lane* on the road (two positions on a lane along the lane direction), and instructs

Table 4. Prompts for Keyword-Based Scenario Program Generation

Step	Sample of prompt patterns
1	Your task is to convert the given scenario semantics into the corresponding representation.
2	Below is an example of scenario semantics and the corresponding scenario representation: <An example of scenario semantics>, <Scenario program example>
3	Below are the definitions and details of elements in the scenario representation: <Details of DSL>, <Details of optional element lists>
4	The oracle in scenario representation is defined by lateral oracle according to ego vehicle behavior.

LMM that the ego vehicle's initial position and destination should perform the ego task in scenario semantics on the road. The prompt pattern is given in Table 3 (2).

- (4) *Divide the road*: CRISER divides the lanes of the road into a group of divisions relative to ego vehicle's initial position, and informs the LMM of the divisions. The prompt pattern of this step is given in Table 3 (3).
- (5) *Guide LMM to generate trajectories for NPC vehicles and pedestrians (for Stage 3 and Stage 4)*: Based on the road divisions, CRISER guides the LMM to generate trajectories of participants that perform corresponding behaviors specified in the abstract scenario. The few-shot for this step is generated by an example trajectory on the road for a basic participant behavior *follow lane ahead* (a trajectory in front along the lane direction on the lane of ego vehicle), to make the LMM learn how to generate the trajectory according to each participant described in the abstract scenario. This prompt pattern is given in Table 3 (4).
- (6) *Add the scenario assertion (for Stage 5)*: CRISER provides the assertion template for the LMM, and prompts it to complete the assertion statements in scenario program. The assertion uses STL to monitor ego vehicle: (i) whether it collides with other objects; (ii) whether it keeps a safe distance from NPC vehicles and pedestrians; and (iii) whether it arrives at destination.

The process of CRISER's keyword-based scenario program generation is given in Table 4, which are referenced to the prompt template in TARGET [23].

4.1.4 Scenario Inspection and Validation. Since LMMs are prone to hallucination, it is uncertain whether the generated intermediate representations of scenario programs are faithfully consistent to the semantics of the original accident-free traffic scenario. A critical challenge in LMM-based scenario program generation lies in verifying the correctness of the CoT process of CRISER.

For the generated waypoint-based scenario programs, CRISER incorporates a specification-based detection to ensure that the CoT-based generation output adheres to the semantics extracted from the original traffic scenario.

- The detection of road statement, ego driving task statement, NPC vehicle statement and pedestrian statement: We detail them in Section 4.2.3, which are also used to ensure the semantic equivalence of the test scenarios generated by the dual-layer exploration of CRISER.
- *The detection of spatial feasibility among NPC vehicles*: The positions of vehicles on the same lane are constrained to be at least 5 meters away from each other at any moment.
- *The detection of spatial feasibility among pedestrians*: The positions of pedestrians are constrained to be unoverlapped at any moment.

For the generated keyword-based scenario programs, CRISER conducts the semantic consistency detection by the lexical analysis. Specifically, CRISER reads the character stream from the scenario program and converts it into the token stream. Then CRISER parses it into a stream of lexical units. The lexical unit is composed of unit type and unit value, which can be divided into five

categories: keywords, identifiers, constants, operators, and delimiters. Based on the stream of lexical units, CRISER constructs the syntax tree and extracts object entities and their hierarchical attribute relationships from the syntax tree, thereby establishing the ownership and dependency between scenario objects (represented as the four-tuple including object type, behavior, position reference and position relation). Then by identifying the keywords of these tuples, CRISER checks whether they are consistent with the scenario semantics of the accident-free traffic scenario.

For the LMM-generated scenario program inconsistent with the original scenario semantics, CRISER generates the feedback to the LMM by pinpointing the elements that violate the scenario semantics and the corresponding participant. The prompt template used for this feedback is:

The generated scenario program does not fully adhere to the original scenario semantics. Specifically, the <element> of the <attribute ID> does not conform to the value of <attribute ID> in the given scenario semantics. Please regenerate the description for the <element> of the <attribute ID> in the scenario program that aligns with the given scenario semantics.

4.2 Specification-Based Dual-Layer Exploration for Safety-Critical Semantic Equivalent Scenarios

Based on the test scenarios generated from traffic videos, CRISER executes them and feeds the ground truth data of test scenarios into the decision-making module of the ADS, testing the driving performance of the ego vehicle during scenario execution. To explore the differences between ego vehicle and human-driving in accident-free traffic scenarios, CRISER uses the dual-layer exploration to search for safety-critical semantic equivalent scenarios and explore their universality, while keeping the semantics of test scenarios consistent with the original accident-free traffic video.

- *Ego-human behavior difference search (outer-layer)*: CRISER refines the scenario space to maximize the differences between ego vehicle’s behaviors and human-driving behaviors for the same driving task in semantic equivalent test scenarios.
- *Safety-violation universality exploration (inner-layer)*: For the discovered SV of the ADS, CRISER explores variations in participants’ trajectories to verify whether the ADS’s safety violation is universal in the traffic situation of the semantic equivalent scenario.
- *Scenario semantic equivalence maintaining*: During the dual-layer search, CRISER describes specifications of scenario semantics and conducts semantic verification on the generated test scenarios, to ensure they preserve the original scenario semantics of the traffic video.

During the dual-layer exploration, CRISER keeps the foundational semantic equivalence of initial conditions and dynamic participant behaviors between the generated test scenarios with the original traffic scenario, which enforces the consistency of dynamic participant behaviors relative to the ego vehicle, but does not strictly constrain interactions between participants. This is because overly preserving interaction semantics (e.g., multi-agent interactions, vehicle-to-vehicle coordination) would limit the diversity of generated scenarios and reduce the effectiveness of exploring safety-critical scenarios.

4.2.1 Ego-Human Behavior Difference Search. The ego-human behavior difference search evolves test scenarios by maximizing the difference between human-driving behaviors in real-traffic scenarios and ego vehicles’ behaviors in semantic equivalent scenarios. Specifically, CRISER employs the feedback-guided fuzzer to search for the test scenarios where the behavior of ego vehicle are significantly different from that of human-driving in semantic equivalent scenarios. The fuzzer is employed based on the fitness of ego-human behavior difference and ego vehicle state deviation.

Ego-Human Behavior Difference Fitness. The primary objective of the outer-layer search is to maximize the high-level behavior differences between human-driving in the traffic scenario and

ego vehicle in semantic equivalent scenarios. The ego-human behavior difference fitness is for the primary objective, which dominates the outer-layer search process. While the secondary objective, ego vehicle state deviation, is considered when the primary objective cannot provide discrimination among candidate scenarios. The selected test scenarios undergo the Gaussian mutation [50] of the participants' positions and speeds to generate new test scenarios. The calculation of the ego-human behavior difference fitness is detailed as follows.

CRISER represents each of their behaviors as an action sequence. The action sequence of human-driving A_h is provided by the scenario semantics of the original traffic video. The action sequence of the ego vehicle A_E is extracted based on the ego vehicle's actual driving trajectory during the execution of the semantic equivalent scenario. CRISER identifies its action sequence by motion state change (mentioned in Section 4.1.1) and action specifications mentioned in Section 4.2.3. CRISER assesses the difference of A_E and A_h by:

$$D(A_E, A_h) = \sum_{i=1, j=1}^{m, n} D(i, j), \quad A_E = \{a_1, a_2, \dots, a_m\}, \quad A_h = \{b_1, b_2, \dots, b_n\},$$

where a_m or b_n is a behavioral action (e.g., *change left*, *turn right*). $D(i, j)$ is calculated by the *Levenshtein Distance*. In the test scenarios and corresponding traffic video, the number of actions in human-driving sequence and ego vehicle sequence may be different and overlapped. The Levenshtein Distance calculates the difference between the two sequences by transforming one sequence into another through insertion, deletion, and replacement operations. For $A_h = \{b_1, b_2, \dots, b_n\}$ and $A_E = \{a_1, a_2, \dots, a_m\}$, the distance calculation adopts a recursive method as follows:

$$D(i, j) = \begin{cases} 0, & i = 0, j = 0 \\ D(i-1, j-1), & a_i = b_j \\ \min \begin{cases} D(i-1, j) + \text{cost_del}(a_i), \\ D(i, j-1) + \text{cost_ins}(b_j), \\ D(i-1, j-1) + \text{cost_rep}(a_i, b_j), \end{cases} & a_i \neq b_j \end{cases}$$

where $\text{cost_del}(a_m)$ represents the cost of deletion operation that deletes a_m from the action sequence A_E . $\text{cost_ins}(b_n)$ represents the cost of insertion operation that inserts a_n into the action sequence A_E . $\text{cost_rep}(a_m, b_n)$ represents the cost of replacement operation that replaces a_m with b_n in the action sequence A_E . For example, $A_h = \{\text{follow lane, decelerate, change right, accelerate, cross}\}$ and $A_E = \{\text{follow lane, brake, change right, accelerate, decelerate, cross}\}$. $D(h, E) = \text{cost_rep}(\text{brake, decelerate}) + \text{cost_ins}(\text{decelerate})$. The cost of insertion and deletion is defined as a constant λ , and the cost of replacement is determined by the types of a_m and b_n . For instance, the cost of replacing a_m and b_n is low if they are *change left* and *change right*, and high if they are *change left* and *brake*. (A_h represents the action sequence of human-driving and A_E represents the action sequence of ego vehicle in the semantic equivalent scenario.)

CRISER adopts two fitness for the ego vehicle state deviation objective: ego vehicle's trajectory deviation and ego vehicle's velocity deviation.

Ego Vehicle's Trajectory Deviation. For the scenario S_i that is evolved from S_{i-1} , the ego vehicle's trajectory deviation is measured by the deviation between the ego vehicle's relative trajectory in S_i and S_{i-1} , whose fitness $Fr(S_i)$ is calculated as below:

$$Fr(S_i) = \max_{t=1}^n \sqrt{(S_i.x_t - S_{i-1}.x_t)^2 + (S_i.y_t - S_{i-1}.y_t)^2}, (S_i.x_t, S_i.y_t) \in \mathbb{P}_i, (S_{i-1}.x_t, S_{i-1}.y_t) \in \mathbb{P}_{i-1}. \quad (3)$$

$(S_i.x_t, S_i.y_t)$ is the ego vehicle's relative position at time t in S_i , where the coordinate system $\mathbb{P}_i$ is defined with the lower-left point of the entrance of ego vehicle's starting lane as the origin. CRISER



Fig. 4. The illustration of a SV S_1 discovered by CRISER's outer-layer exploration.

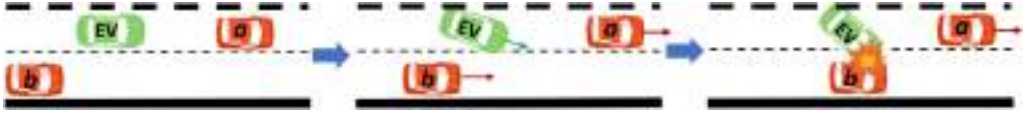


Fig. 5. The illustration of a variant scenario of S_1 generated by CRISER's inner-layer exploration.



Fig. 6. The illustration of a SV S_2 discovered by CRISER's outer-layer exploration.

sets the initial position of ego vehicle at the entrance of its starting lane, thus $\mathbb{P}_i.0 = \mathbb{P}_{i-1}.0$. A higher $Fr(S_i)$ indicates a larger relative trajectory deviation.

Ego Vehicle's Velocity Deviation. Let ego vehicle's velocity at time t in scenario S_i be $\mathbf{v}_i(t) = [v_{i,x}(t), v_{i,y}(t)]$, with its speed value and heading angle defined, respectively, as $v_i(t)$ and $\theta_i(t)$:

$$v_i(t) = \|\mathbf{v}_i(t)\|, \quad \theta_i(t) = \text{atan2}(v_{i,y}(t), v_{i,x}(t)). \quad (4)$$

The fitness functions for ego vehicle's velocity deviation of S_i include velocity magnitude difference $D_v^m(S_i)$ and velocity direction difference $D_v^a(S_i)$:

$$D_v^m(S_i) = \max_{t=1}^N |v_i(t) - v_{i-1}(t)|, \quad D_v^a(S_i) = 1 - \frac{1}{N} \int_1^N \frac{\mathbf{v}_i(t) \cdot \mathbf{v}_{i-1}(t)}{v_i(t) \cdot v_{i-1}(t)} dt. \quad (5)$$

A higher $D_v^m(S_i)$ or $D_v^a(S_i)$ indicates a greater ego vehicle's velocity deviation.

When the objective of ego vehicle state deviation is applied, CRISER selects the Pareto-optimal test cases based on their values across the two fitness of the objective. Specifically, all of the test cases that exhibit superior performance in at least one objective are retained within the population.

4.2.2 Safety-Violation Universality Exploration. For each SV discovered by *ego-human behavior difference search*, CRISER incrementally explores variations into the concrete parameter values of participants' attributes in the SV (within the semantics of the original scenario). If the safety violation of the ADS persists across diverse variations of the semantic equivalent scenarios, it is considered as a potentially universal deficiency of the ADS's decision-making.

Here we provide two examples to better illustrate the safety-violation universality validation. Figure 4 shows a SV discovered by CRISER's outer-layer exploration. The inner-layer exploration of CRISER applies perturbations to the initial speeds of NPC vehicles and the initial distances between the ego vehicle and two NPC vehicles. An example of the variant scenario is given as Figure 5. Across multiple generated variants, the same safety violation consistently reoccurs. Therefore, this SV is considered universal. In contrast, for another SV discovered by CRISER's outer-layer exploration, shown as Figure 6, the safety violation no longer occurs in most variants (an example of variant is shown as Figure 7). This demonstrates that the discovered safety violation is an extreme or accidental case.



Fig. 7. The illustration of a variant scenario of S_2 generated by CRISER's inner-layer exploration.

The diversity between the SV and its variant is calculated as follow. For the SV sf and the set of its variations $\mathbb{Z}$ where the safety violation of the ego vehicle occurs, the range of variations RV_{sf} is evaluated by the maximal *Euclidean distance* between participants' trajectories across sf and a scenario in $\mathbb{Z}$. Euclidean distance is widely used to compute the similarity of participant trajectories [36] across two scenarios:

$$RV_{sf} = \arg \max_s \sum_{s \in \mathbb{Z}} ED_{sf,s}, \quad ED_{sf,s} = \frac{\sum_{n=1}^l \sum_{m=1}^c TD_{sf^n,s^m}}{l * c}, \quad (6)$$

where ED represents the Euclidean distances of two test scenarios and TD represents the Euclidean distances of two participant trajectories, calculated as follows:

$$TD_{s_i^n,s_j^m} = \sum_{k=0}^{\mu} \sqrt{(x_{s_i^n,k} - x_{s_j^m,k})^2 + (y_{s_i^n,k} - y_{s_j^m,k})^2}. \quad (7)$$

sf^n represents the n th participant trajectory in sf . The number of NPC vehicles/pedestrians in sf and s are l and c respectively. $(x_{sf^n,k}, y_{sf^n,k})$ represents the position of the k th waypoint of the n th participant trajectory in sf . μ represents the number of waypoints in the participant trajectory sf_i . If $RV_{sf} \geq M$, CRISER records these SVs which can be replayed to reproduce the safety violation of the ADS.

4.2.3 Scenario Semantic Equivalence Maintaining. To keep the initial conditions and dynamic participant behaviors of generated test scenarios semantically equivalent to those of the original accident-free traffic scenario, CRISER describes the specifications for them to check the semantics equivalence between the original traffic scenario with each generated test scenario. For the generated test scenario where any attribute violates the specifications, CRISER re-generates it.

According to the extracted scenario semantics from traffic videos, the specifications for semantics equivalence are divided into three groups: road topology, ego vehicle, initial conditions, and behaviors of dynamic participants.

Road Topology. The road topology of the original traffic scenario is represented as $\mathbb{R}_o = (T, N, \vec{D}, \mathbb{G})$. T is the road type, N is the number of lanes on the road, $\vec{D} = (df(l_1), \dots, df(l_N))$ are the directions of these lanes. $\mathbb{G} = (g_1, \dots, g_M)$ are the traffic signs on the road. According to the map and coordinates declared in the test scenario program, CRISER retrieves the road topology in the test scenario from the map file, represented as $\mathbb{R}_s = (T', N', \vec{D}', \mathbb{G}')$. The specifications for semantic equivalence of road topology are:

$$(T' = T) \wedge (N' = N) \wedge (\forall i \in [1, N], df(l'_i) = df(l_i)) \wedge (\forall j \in [1, M], g_j = g'_j).$$

Ego Vehicle. The starting point and ending point of the camera vehicle (CV, the vehicle with the dashcam recording the traffic video) are represented as S and D respectively. The initial position and destination of ego vehicle in the test scenario are represented as I and F . The specifications for the semantic equivalence of ego vehicle are:

$$l_i, l_j \in \mathbb{R}_o, l'_i, l'_j \in \mathbb{R}_s, (S \in l_i, D \in l_j) \wedge (I \in l'_i, F \in l'_j).$$

Initial Conditions and Behaviors of Dynamic Participants. The specifications include behavior specifications and relative position specifications. The behavior specifications are divided into lane specifications, waypoint specifications, direction specifications, and speed specifications. We describe them respectively as follows.

The movement of each dynamic participant is defined by a trajectory in the test scenario program. Each trajectory consists of a sequence of waypoints, and each waypoint $w_i = (pos^i, vel^i)$ contains the position ($pos^i = (x^i, y^i)$) and velocity ($vel^i = (v_x^i, v_y^i)$). For the defined trajectory of each participant, CRISER splits it into a sequence of segments using linear interpolation on the trajectory to identify its significant changes of motion states (mentioned in Section 4.1.1). For the sequence of trajectory segments, CRISER uses behavior specifications to sequentially check whether their semantics are equivalent to the behavior sequence of the participant in the original traffic scenario. Here we take the specifications of *change left* (the relative position is *right front*) as the example, and the others can be found in Section A in Appendix A.

- *Lane specification for change left:* This specification verifies that the starting position and ending position are on two different lanes on the same road, and the directions of two lanes are the same. The ending position is on the left lane of starting position. pos^0 is the position of the initial waypoint on the trajectory and pos^F is the last waypoint. $fd(A, B)$ is the direction from A to B:

$$pos^0 \in l_s, pos^F \in l_f, (s \neq f) \wedge (l_s, l_f \in \mathbb{R}_s), \\ (df(l_s) = df(l_f)) \wedge (-1 < \sin(fd(pos^0, pos^F), df(l_s)) < 0).$$

- *Waypoint specification for change left:* This specification verifies that the positions of intermediate waypoints of the trajectory segment are between the starting and ending positions. i represents the index of the waypoint:

$$\forall i \in (0, F), pos^i \in l_s \cup l_f, \\ pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], \\ pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)].$$

- *Direction specification for change left:* This specification verifies that the driving directions are to the left:

$$\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (90, 180).$$

- *Speed specification:* $speed_{max}$ is the road speed limit, and h_c is the threshold for speed change during the actions except accelerating, decelerating, and braking:

$$\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1} - vel^i|/\Delta t < h_c).$$

- *Relative position specification for right front:* This specification verifies the initial position of the participant's behavior is in the right front of the initial position I of ego vehicle:

$$0 < \sin(fd(I, pos^0), d(l_i')) < 1, \quad I \in l_i'.$$

5 Evaluation

To demonstrate the ability of CRISER [17], we apply it to test an industrial-grade full-stack ADS, Apollo 7.0 [8], which is widely used to control AVs on public roads. To evaluate the efficiency and effectiveness of CRISER, we answer the following **Research Questions (RQs)**:

- *RQ1:* How effective is CRISER's traffic video-based scenario understanding and generation?
- *RQ2:* Can CRISER keep the semantics of the generated test scenarios equivalent to the original traffic video?

- *RQ3*: Can CRISER discover safety violations of Apollo's decision-making in semantic equivalent scenarios from accident-free traffic videos?

5.1 Experiment Settings

5.1.1 ADS under Test. We target the safety testing of industry-grade L4 ADS. In particular, we select the industrial open source ADS, Baidu Apollo [8], for the following reasons. (1) *Representativeness*. Apollo community ranks among the top-four leading industrial ADS developers [5], while the other three are not publicly released. (2) *Practicality*. Apollo has been commercialized for many real-world self-driving services [2, 4]. Our proposed method can be applied to test other ADSs as well.

5.1.2 Experimental Environment. We conducted the experiments on Ubuntu 20.04 with 500 GB memory, an Intel Core i7 CPU, and an NVIDIA GTX2080 TI. SORA-SVL [10] (an AV simulation platform that supports connection with Apollo) is selected to execute the generated scenarios.

5.1.3 Dataset. We select the automotive video dataset, **HRI Driving Dataset (HDD)** [42], to generate traffic-realistic scenarios. HDD is a dataset to reflect driver behavior and various driving situations in real traffic, which includes 104 hours of safe human driving in the San Francisco collected using an on-vehicle recorder. The dataset encompasses various types of roads (straight road, intersection, T-junction) and participant behaviors (follow lane, change left/right, turn left/right, cross, accelerate, decelerate, brake, walk along, walk across, and stand). We adopt GPT-4V [13] as the LMM because it is the current SOTA LMM, widely known and easily accessible.

5.1.4 Parameter Settings. Some parameters of CRISER's motion change-based key frame extraction require to be defined: the interpolation parameter α , the coefficient β of the intermediate vector deviation threshold τ , the minimum motion threshold b_f of the background flow magnitude for slow movers, and the angle threshold b_r of new flow direction deviation for occlusions. The parameters above are determined through grid-based sensitivity analysis using 50 scenario videos selected from the HDD dataset. Each scenario contains parked vehicles on both sides of the road while pedestrians cross the road. Such scenarios are intentionally selected because they combine the three major sources of difficulties for optical-flow-based motion analysis: occlusion, slow movers, and camera movement. Based on these scenario videos, we conduct a series of joint configuration experiments of the parameter sensitivity analysis for CRISER's scenario video information extraction pipeline, and the results are given in Table B1 in Appendix B.

α is set as 0.2, as the key action extraction of CRISER is required to identify and retain the instantaneous changes of motion changes. The smaller value of α is sensitive to sudden changes and can more promptly reflect changes in actions. β is set as 1.5, which achieves the highest F1-score, offering the best tradeoff between false positives and missed detections. b_f is set as 2%, which preserves slow movers without amplifying background noise. b_r is set as 30, which ensures the identification of true motion continuity while reducing errors from false trajectory links. Besides the parameters in CRISER's scenario video information extraction, the threshold h_c in the speed specification is set as 0.5 m/s^2 due to that: according to research on driving behavior [40, 46], changes in speed below 0.5 m/s^2 during human driving are often considered steady and not recognized as obvious acceleration or deceleration action. We set M as 20 referencing to [36, 52], which indicates that different test scenarios with Euclidean distance greater than 20 can be used as classification criteria for different categories of scenarios.

5.2 Experiment Design

For RQ1, we use CRISER to extract the scenario semantics of traffic videos, and then generate test scenario programs. To evaluate the effectiveness of scenario semantics extraction for traffic videos, we employ M-CPS [56] and $CRISER_E$ as the baselines. M-CPS is a SOTA model to extract effective information from accident videos. $CRISER_E$ is a variation of CRISER for ablation evaluation of CRISER's motion change-based key framework extraction. $CRISER_E$ directly splits the traffic video at 1 s timestep and leverages GPT-4V to understand the semantics of the sequences of frames. We randomly select 100 different videos that encompass various types of roads, ego vehicle's driving tasks, participant types and behaviors. Then we run CRISER, M-CPS and $CRISER_E$ to extract scenario semantics of the same videos, respectively.

To validate the effectiveness of CRISER in test scenario generation, we use CRISCO [3] and $CRISER_D$ as the baselines. CRISCO is a SOTA technique that defines and solves a large set of constraints to generate test scenarios based on the given scenario semantics. $CRISER_D$ is a variation of CRISER for ablation evaluation of CRISER's CoT-empowered test scenario generation. $CRISER_D$ directly leverages GPT-4V to generate test scenarios (only providing the first-step prompt and example of scenario program with parameter introduction) according to the scenario semantics. Next, we run CRISER, CRISCO and $CRISER_D$ to generate test scenarios based on the same scenario semantics. Note that we run CRISER and $CRISER_D$ to generate both waypoint-based scenario programs and keyword-based scenario programs of the scenario semantics, respectively. Here we select AVUnit [3] as the DSL for waypoint-based scenario program description, and TARGET [23] as the DSL for keyword-based scenario program description.

Four of the authors independently analyze and cross-check the extracted semantics of each scenario video and the generated scenario programs of each scenario semantics. Another author is involved in a group discussion to resolve conflicts and reach agreements.

CRISER utilizes LMM's understanding and reasoning capabilities to interpret traffic videos and generate test scenarios. We further evaluate its applicability over other LMMs. In our experiments, CRISER uses two other LMMs, Llama-3.2-11B-Vision [16] and Gemma-3-1b-pt [12], to understand the same scenarios and generate waypoint-based and keyword-based scenario programs.

For RQ2 and RQ3, we randomly select 100 traffic videos from HDD dataset, including scenarios on straight roads, curved roads, crossroads and T-junctions. Then we use CRISER to generate test scenarios to test Apollo. To answer RQ2, for each generated test scenario program, we use the LMMs (including GPT-4V, Llama-3.2-11B-Vision, and Gemma-3-1b-pt) to identify whether the semantics of the test scenario are equivalent to the semantics of the original traffic video.

To validate the semantic equivalence maintaining in CRISER, we conduct a user study with domain experts to complement the evaluation on the semantic equivalence of the test scenarios generated by the CoT-empowered LMM's generation and the dual exploration of CRISER to the original traffic video semantics.

To answer RQ3, for the SVs discovered by CRISER, we analyze the potential deficiencies and correct operations of modules in Apollo. To evaluate the effectiveness and efficiency of CRISER in discovering safety violations of the ADS in semantic equivalent scenarios, we employ three SOTA approaches that can generate and evolve test scenario programs as baselines, M-CPS [56], LeGEND [51], and SoVAR [32]. M-CPS builds test scenarios by reproducing and mutating scenarios from traffic videos. LeGEND can evolve the given test scenario programs from multiple objectives. SoVAR can fuzz the test scenario programs by the generalizable constraint solver. Besides, we also employ a variant of CRISER, $CRISER_R$, for the ablation evaluation of CRISER's dual-layer safety-critical scenario exploration. $CRISER_R$ replaces the dual-layer exploration of CRISER by randomly searching participants' parameters within their spaces while preserving the semantic equivalence

maintaining in CRISER. Furthermore, to evaluate the role of maintaining the semantic equivalence of test scenarios' initial conditions and participant behaviors, we employ another variant of CRISER, *CRISER_S*, which removes the semantic equivalence maintaining from the dual-layer exploration of CRISER. We run CRISER and the above baselines based on the same scenario programs of accident-free traffic videos to generate the same amount of test scenarios, and compare from the following aspects:

- How many types of Apollo safety violations are found by CRISER and baselines in their generated test scenarios?
- How many different traffic videos are leveraged by CRISER and baselines to expose safety violations of Apollo?

To evaluate the generalization of our findings, we execute the SVs discovered by CRISER to test another open source ADS, Autoware [18], within another simulator Carla [24].

To account for randomness, the experiments of the above three RQs are repeated five times.

5.3 Result Analysis: RQ1

For the accuracy of scenario semantics extraction from traffic videos, we evaluate it by calculating the information extraction accuracy of elements in scenarios. The elements are divided into four categories: road, ego driving task, participants, and relative positions. Each category of elements include several attributes. The attributes of road include road type and traffic signal. The attributes of ego driving task include the behaviors of ego vehicle. The attributes of participant include participant types and participant behaviors. The attributes of relative positions include the initial positions and destinations of participants relative to ego vehicle. Here the understanding of an element in a scenario is accurate if all attributes of the element are extracted correctly. The information extraction accuracy of the elements of category c is represented as SUA_c , calculated as follows:

$$SUA_c = \frac{1}{\|\mathbb{TV}\| \times \|A_i^c\|} \sum_{i=1}^{\|\mathbb{TV}\|} \sum_{j=1}^{\|A_i^c\|} \mathbf{1} \left(\bigwedge_{ar \in A_i^c.j} M(ar, \mathbb{TV}.i) \right).$$

where $\mathbb{TV}$ represents the set of traffic scenarios and A_i^c represents all extracted elements of category c from i th scenario. $\mathbf{1}$ is an indicator function mapping a Boolean condition to a value in $\{0, 1\}$: if the condition is true, it returns 1; otherwise, it returns 0. $\bigwedge$ means logical AND. $A_i^c.j$ represents the extracted attribute values of the j th element of category c in the i th scenario. M is a function that evaluates whether the extracted value of the attribute conforms to the ground truth of corresponding traffic scenario.

Figure 8 shows the accuracy of CRISER, the ablated variant *CRISER_E*, and the baseline approach M-CPS, in extracting scenario semantics from traffic videos. On the extraction of most categories of scenario elements, the accuracy of CRISER is high and performs better than them. As shown in Figure 8(1), the relatively low accuracy of *CRISER_E* demonstrates the effectiveness of the motion change-based key frame extraction of CRISER. As shown in Figure 8(2), M-CPS performs well in extracting static information (e.g., road structure) from traffic scenarios. However, for the information of complex dynamic semantics (e.g., participant behaviors), the extraction accuracy of M-CPS significantly decreases. The same problem also exists for CRISER, but the decrease is slight and it can almost accurately extract key information of dynamic semantics.

For the correctness of test scenario generation, we use **Test Scenario Correctness (TSC)** as the metric. Here a test scenario is correct if all elements during the execution of the scenario conform

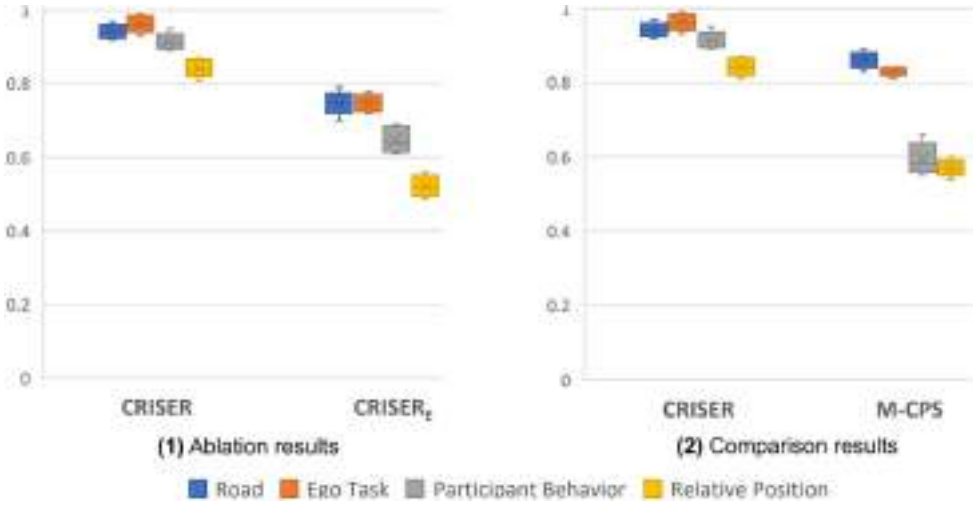


Fig. 8. Scenario semantics extraction accuracy.

to the scenario semantics of corresponding traffic video. The calculation of TSC is given as follows:

$$\text{TSC} = \frac{1}{\|\mathbb{S}\|} \sum_{i=1}^{\|\mathbb{S}\|} \mathbf{1} \left(\bigwedge_{sr \in \mathbb{S}.i} \text{Sim}(sr, \mathbb{AS}.i) \right),$$

where $\mathbb{S}$ represents the set of generated test scenarios and $\mathbb{AS}$ represents the set of scenario semantics. sr represents the execution result of an element in the concrete scenario. Sim is to determine that during the execution of test scenario, whether the element in it conforms to the scenario semantics.

Due to the varying complexity of scenarios on different road types, there are differences in the correctness of test scenario generation. We conduct statistical analysis on the correctness of test scenario generation. Figure 9 shows the results, where CRISER:w and $\text{CRISER}_D:w$ represent the correctness of CRISER and CRISER_D in generating waypoint-based scenario programs for scenario semantics, and CRISER:k and $\text{CRISER}_D:k$ represent the correctness of CRISER and CRISER_D in generating keyword-based scenario programs for scenario semantics. As shown in Figure 9(1), compared to the ablated variant CRISER_D , CRISER's correctness in generating waypoint-based scenario programs is significantly higher, which demonstrates the effectiveness of CRISER's CoT-empowered LMM-based scenario program generation according to given scenario semantics. As shown in Figure 9(2), the correctness of CRISER is also higher than that of the baseline (CRISCO) especially at T-junction, which demonstrates the effectiveness of the multi-modal few-shot CoT that guides the LMM to generate waypoint-based scenario programs according to scenario semantics.

From Figure 9, it can be seen that the correctness of both CRISER and CRISER_D in generating keyword-based scenario programs are higher than in generating waypoint-based scenario programs. This performance gap is primarily due to the higher complexity of waypoint-based scenario programs. Despite this, the waypoint-based scenario program is fundamental to the downstream scenario evolution and exploration in CRISER. The waypoint-based scenario programs are more feasible and convenient for customizing and exploring the safety-critical cases of the accident-free traffic situations to discover potential decision-making safety violations in the ADS.

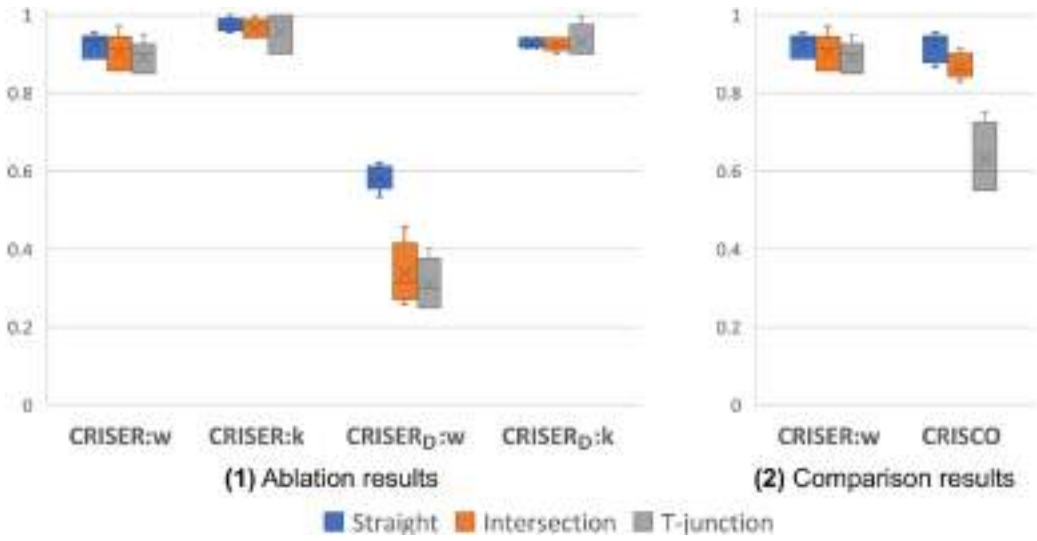


Fig. 9. Test scenarios generation correctness.

Table 5. Scenario Semantics Extraction Accuracy

Element category	Road	Ego task	Participant	Relative positions
CRISER	97%	96%	95.04%	86.18%
CRISER _A	94%	92%	91.32%	85.21%
CRISER _G	90%	87%	81.82%	79.05%

Table 6. Scenario Generation Correctness

Road type	Straight	Intersec	T-junc
CRISER:w	95.56%	91.43%	90%
CRISER:k	97.78%	97.14%	95%
CRISER _A :w	91.11%	82.86%	80%
CRISER _A :k	95.56%	91.43%	90%
CRISER _G :w	84.44%	74.29%	75%
CRISER _G :k	93.33%	85.71%	85%

We select two alternative LMMs, Llama-3.2-11B-Vision [16] and Gemma-3-1b-pt [12], as the replacement of GPT-4V in CRISER. This gives two variants: *CRISER_A* and *CRISER_G*. Tables 5 and 6 show the results of CRISER and variants. In Table 6 (Intersec and T-junc represent intersection and T-junction), CRISER:w, *CRISER_A*:w and *CRISER_G*:w represent the correctness of CRISER, *CRISER_A* and *CRISER_G* in generating waypoint-based scenario programs, and CRISER:k, *CRISER_A*:k and *CRISER_G*:k represent the correctness of CRISER, *CRISER_A* and *CRISER_G* in generating keyword-based scenario programs. Their accuracy and correctness in traffic video scenario semantics extraction and test scenario generation remain largely stable, with two variants exhibiting slight declines. The results indicate that CRISER can be generalized to different LMMs with similar functionalities.

5.4 Result Analysis: RQ2

In each run, CRISER generates 10 test scenarios for each of the 100 traffic videos. Considering that the CoT-empowered LMMs can accurately generate test scenario programs according to scenario semantics, we use the similar CoT prompting pattern to guide the LMM to identify whether the semantics of the 1,000 generated test scenarios are equivalent to the semantics of corresponding traffic videos. The CoT prompts are given as Table 7.

To test whether the CoT-empowered LMM can identify the semantic equivalence between generated test scenarios and the original scenarios of traffic videos correctly, we randomly select 100 test scenarios generated by the dual-layer exploration of CRISER, and guide each of the three LMMs to identify their semantic equivalence to the corresponding original scenarios of traffic

Table 7. Prompts for Semantics Identification of Test Scenario Programs

Type	Sample of prompt patterns
(1)	The given table is the semantics of the original scenario on the given road. The <i>original_program</i> is the scenario program of the original scenario.
(2)	The scenario program contains five parts: road statement, ego driving task statement NPC vehicles' trajectories definition, pedestrians' trajectories definition, scenario assertion statement.
(3)	For the <i>test_scenario_program</i> , please identify its semantics and assert whether its semantics are equivalent to the semantics of original scenario.

Table 8. Test Scenario Semantic Equivalence Identification Correctness of the CoT-Empowered LMM

	1	2	3	4	5
GPT	96%	94%	91%	97%	92%
Llama	93%	92%	92%	90%	91%
Gemma	90%	94%	93%	95%	91%

Table 9. Semantic Equivalent Scenarios in 1,000 Generated Test Scenarios

	1	2	3	4	5
GPT	984	991	989	990	993
Llama	988	992	990	991	995
Gemma	990	994	992	993	996

videos. To check the correctness of CoT-empowered LMM in identifying test scenario semantics, four of the authors independently analyze and cross-check each identification result of the test scenario's semantic equivalence to the corresponding semantics of the original accident-free traffic video. Another author is involved in a group discussion to resolve conflicts and reach agreements. The correctness of CoT-empowered LMM in identifying semantic equivalence of generated test scenarios is given in Table 8. From the results, it can be seen that in each run of test, the CoT-empowered LMM can almost correctly identify the semantic equivalence between generated test scenarios with original accident-free traffic scenarios.

For the 1,000 test scenarios generated by the dual-layer exploration of CRISER, the results of each LMM's identifying their semantic equivalence to corresponding scenarios of original traffic videos are given in Table 9. From the results, it can be seen that in each run, more than 980 in the 1,000 generated test scenarios are identified as semantic equivalent to the corresponding accident-free traffic videos by the three LMMs. The results can demonstrate that the specification-based dual-layer exploration of CRISER can keep the semantics of generated test scenarios equivalent to the original accident-free traffic video scenarios.

For the user study of the semantic equivalence between the generated test scenarios with the corresponding traffic scenario video, we randomly select 15 accident-free traffic videos, and their test scenarios generated by CRISER's CoT process and CRISER's dual-layer exploration. Then we execute them to test the ADS, and record the execution process. For each of them in the questionnaire, we provided: (1) the original traffic video, (2) a simulation replay video of the test scenario program generated by CRISER's CoT process, and (3) the SV discovered by CRISER from the test scenario (if ADS safety violation is detected in the scenarios evolved from the scenario program). Each participant of the questionnaire is asked to rate the semantic equivalence between the original video and the generated test scenarios on a 5-point scale (1: totally equivalent, 5: totally not equivalent). The questionnaire is designed as google.

We distributed questionnaire invitations to 80 participants from three research universities and two industry-leading companies. All participants held a master's degree or higher in software or automotive engineering, and had no prior collaboration with the authors of this article to ensure unbiased feedback. To gather the subjective responses and ensure the validity of the collected

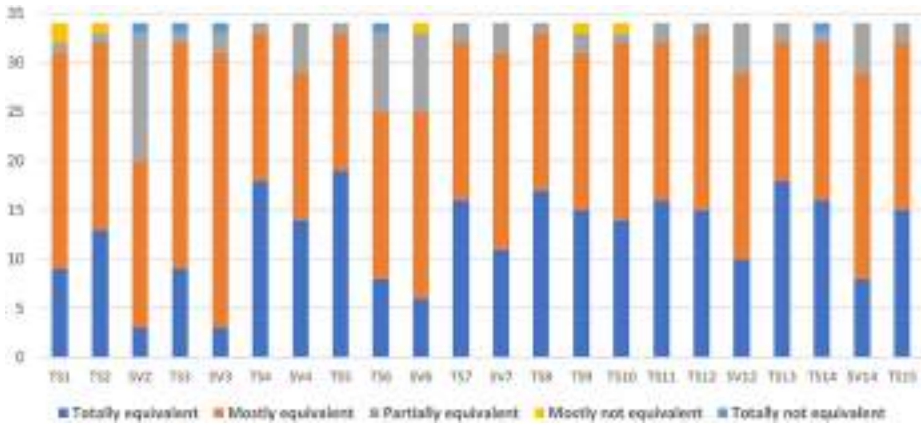


Fig. 10. Results of each question in user study.

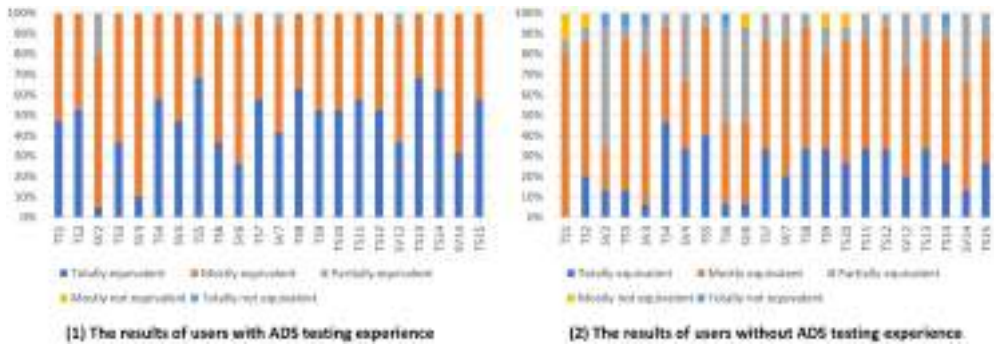


Fig. 11. Result comparison of users with and without ADS testing experience.

response: (1) Participants are required to complete a scenario review task before filling out the questionnaire. (2) Responses are checked for consistency, and incomplete submissions are excluded.

We received 34 valid responses after excluding incomplete or duplicate submissions. Among the valid participants, 19 individuals had prior experience with ADS simulation testing (1 participant with over 5 years of experience, 3 participants with 3–5 years of experience, and 15 participants with 1–2 years of experience). The remaining 15 participants had general software testing expertise but lacked specific experience in ADS simulation test scenario description. The results of the user survey are given in Figures 10 and 11.

According to the questionnaire results shown as Figure 10, for 86% of generated simulation test scenarios, more than 80% of their received scores are rated as *totally equivalent* or *mostly equivalent* to their corresponding original human-driving videos. This result demonstrates that the majority of CRISER's generated simulation test scenarios successfully preserve the fundamental semantics of corresponding traffic scenarios. Among them, for 71% of SVs discovered by CRISER in the survey, more than 80% of their received scores are rated as *totally equivalent* or *mostly equivalent*, suggesting that these violations indeed arise from the initial conditions and participant behaviors of the human-driving scenarios.

We further make a statistical analysis between participants with and without prior ADS simulation testing experience, shown as Figure 11. The comparison results show that the semantic equivalence scores of CRISER's generated scenarios rated by experienced participants are higher

than those rated by inexperienced participants, indicating that users familiar with ADS testing tend to understand and recognize stronger semantic equivalence between generated simulation test scenarios and original human-driving scenarios. For few SVs (e.g., SV2 and SV3), there exists a large discrepancy between participants with and without ADS testing experience. We investigate them and attribute this discrepancy to differences in interpreting the early termination of SVs (SVs). In these SVs, the simulation is terminated early once discovering the safety violation of ego vehicle, in which the behaviors of the ego vehicle and participants have not yet fully executed. Participants without ADS testing experience expect the full execution of the ego vehicle's and all participants' behaviors as described in the original scenario semantics, thus perceiving the incomplete execution due to early termination as a deviation from semantic equivalence.

5.5 Result Analysis: RQ3

In the semantic equivalent test scenarios generated by CRISER, on average, 146 (min 107, max 182) of them expose safety violations of Apollo. Among these SVs, 3 on average (min 2, max 6) of them involve safety violations caused by illegal actions of participants, and we exclude them from the experimental results in the following analysis. The results indicate that CRISER is effective in discovering safety violations of ADS's decision-making in test scenarios semantic equivalent to the original accident-free traffic videos.

To better analyze and clarify the potential deficiencies of the decision-making of Apollo, we classify the discovered safety violations into distinct types by the error of Apollo. The results are shown as Table 10, where EV represents the ego vehicle. NV represents NPC vehicles (2 NVs represent two NPC vehicles), and P represents pedestrians. In represents Intersection, T represents T-junction, St represents Straight Road. As the page limitation, we select one to explain in the following. The illustration for all types of safety violations are provided in Appendix C. Here we take the safety violation 5 as the example: the driving task of EV and human driver is driving along the lane to go through. Vehicle *a* is driving in front of EV on the lane and vehicle *b* is driving in front of EV on the adjacent lane. In the upper scenario of Figure 12, when vehicle *a* stops, EV plans an overtaking trajectory. When EV changes to the lane of vehicle *b*, the vehicle *b* decelerates and stops in the front of EV. The distance between vehicle *b* and EV is enough for EV to change to the left lane to continue the driving task. However, EV fails to adjust a new trajectory to change lane again. In the lower scenario of Figure 12, when EV starts to change lane, vehicle *b* decelerates and stops. EV drives forward for a not short distance, leading to collision with vehicle *a*. For the human-driving in this traffic scenario, it safely changes to the right lane with driving forward for a short distance. When vehicle *b* stops, it changes lane again to the original lane.

In Table 10, Tr/Tl represent turn right/turn left, and Cl/Cr represent change left/change right. Dt represents drive through. rp represents the relative position, where lv/rv is left vertical/right vertical, lf/rf is left front/right front, lb/rb is left behind/right behind, ah is ahead, and op is opposite.

To evaluate the generalization of our findings, we execute the SVs discovered by CRISER to test another open source ADS, Autoware [18] in another simulator Carla [24]. The recurrence rates of discovered types of safety violations on Autoware are given in Table 11.

The 10 types of decision-making SVs discovered by CRISER can be consistently reproduced on Autoware, with reproduction rates exceeding 60% for all types and over 75% for most of them. The results demonstrate that our findings are not confined to a specific ADS implementation or simulation platform, demonstrating the high generalizability and cross-system applicability of CRISER's discovered safety violations.

In each run, CRISER and baselines respectively evolve and generate 10 test scenarios from each of the 100 scenario programs. For the SVs discovered by each approach, we identify the cases that are not caused by the ADS's decision-making (we call them *non-ADS-induced safety violation*): (1)

Table 10. The Distinct Types of Safety Violations of Apollo's Decision-Making Discovered by CRISER

ID	Road type	Ego task	Participant			Root cause summary	Responsible submodule
			type	rp	behaviors		
1	In/T	Tr /Tl	2NV	lv/ rv	cross cross + accelerate	Dynamic behavior and intent prediction: Fail to distinguish the independent kinematic states of continuous vehicles	Prediction
2	St	Cl /Cr	2NV	ah lb/rb	decelerate accelerate	Behavioral hesitancy and trajectory re-planning: Fail to promptly generate trajectory for overtaking and adjustment during ego vehicle's lane-changing	Planning
3	In	Cross	2NV	lv	cross turn left	Interactive behavior prediction: Fail to predict the changing intent of turning maneuver during ego vehicle's crossing	Prediction
4	In/T	Tl	1NV	lv	turn left	Complex right-of-way handling: An inadequacy of dynamic reasoning to correctly assign right-of-way among vehicles on multi-direction lanes in connection area of intersection	Planning
5	St	Dt	2NV	ah lf/rf	stop and decelerate	Multi-obstacle trajectory coordination: Fail to generate and switch to a feasible alternative path in real-time encountering obstacles on multi-lanes	Planning
6	In/T	Tr	1NV	lv	cross and stop	Path re-planning for alternative lane: Fail to activate to utilize valid adjacent lane for alternative path during turning	Planning
7-1	In/T	Tl	2NV	op	cross	Inability of collaborative longitudinal motion modeling among multi-vehicles	Prediction
7-2		Cross	2NV	lv	cross		
8	St	Dt	2NV	ah	decelerate	Oscillating and uncertain trajectory forecasts for extremely slow-moving (less than 1 m/s) vehicles ahead: Inconsistently classify slow-moving vehicles ahead as either static obstacles or dynamic entities with negligible velocity	Prediction
9-1	In/T	Tr	truck	lv	cross	The computed safety margin is insufficient to account for truck's lateral footprint and drift near the lane marking	Planning
9-2		Cross	truck	rv	turn right		
10	In/T	Cross	2NV	ah lv	cross; turn left	Scenario transition awareness and subsequent behavior generation: Fail to transition from crossing maneuver to a proper car-following or lane-changing in a timely manner	Planning

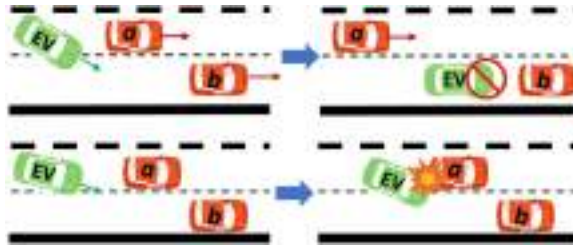


Fig. 12. Two examples of safety violation 5.

Table 11. The Recurrence Rates of Discovered Types of Safety Violations on Autoware

ID of safety violation	1	2	3	4	5	6
Recurrence rate on Autoware	84.62%	88.89%	92.86%	82.35%	90.91%	80%
ID of safety violation	7-1	7-2	8	9-1	9-2	10
Recurrence rate on Autoware	75%	90%	100%	75%	71.43%	63.16%

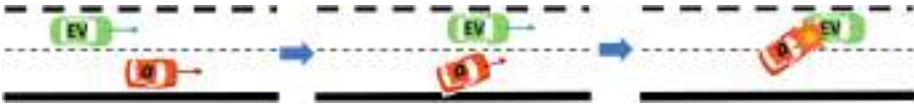


Fig. 13. The first example of safety violation not caused by ego vehicle.

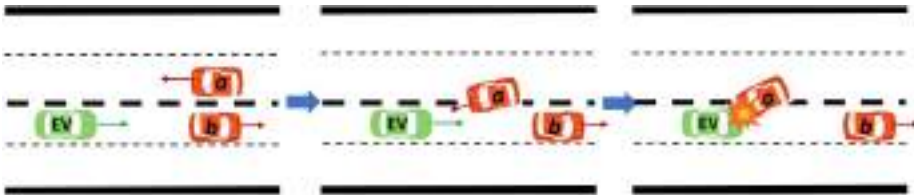


Fig. 14. The second example of safety violation not caused by ego vehicle.

The participant actively collides with the ego vehicle from the side or rear of the ego vehicle. An example is shown as Figure 13, where the NPC vehicle *a* changes to the lane of ego vehicle (with the right-of-way) illegally and directly collides to the rear of the ego vehicle. (2) The NPC vehicle exhibits abnormal behaviors (e.g., retrograding) prior to the occurrence of the safety violation. An example is shown as Figure 14, where the NPC vehicle *a* retrogrades into the lane of the ego vehicle and leads to the collision. (3) The ego vehicle fails to arrive at destination caused by the road block where there's no feasible path for the ego vehicle to arrive at the destination. An example is shown as Figure 15, where the point D represents the destination. In these test scenarios, the resulting safety violations are not attributable to the ADS's decision-making.

The comparison results of CRISER and baselines on discovered types of safety violations are given as Table 12. We exclude the non-ADS-induced SVs in further analysis.

The detailed comparison results are given in Figures 16 and 17. In Figure 16, among the 100 accident-free scenarios, on average, CRISER utilizes 74 (min 66 and max 81) of them to discover 143 (min 105 and max 176) safety violations of Apollo's decision-making. M-CPS utilizes 25 (min 18 and max 31) accident-free scenarios to discover 38 (min 21 and max 55) safety violations of Apollo. LeGEND utilizes 39 (min 26 and max 49) accident-free scenarios to discover 112 (min 81 and max

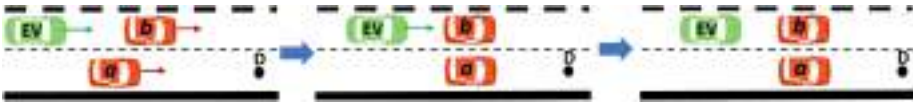


Fig. 15. The third example of safety violation not caused by ego vehicle.

Table 12. The Comparison Results of CRISER and Baselines on Discovered Types of SVs

	CRISER	M-CPS	LeGEND	SoVAR	$CRISER_R$	$CRISER_S$
Safety violation types	10	3	13	4	6	13
Non-ADS-induced safety violation type	0	0	6	0	0	8

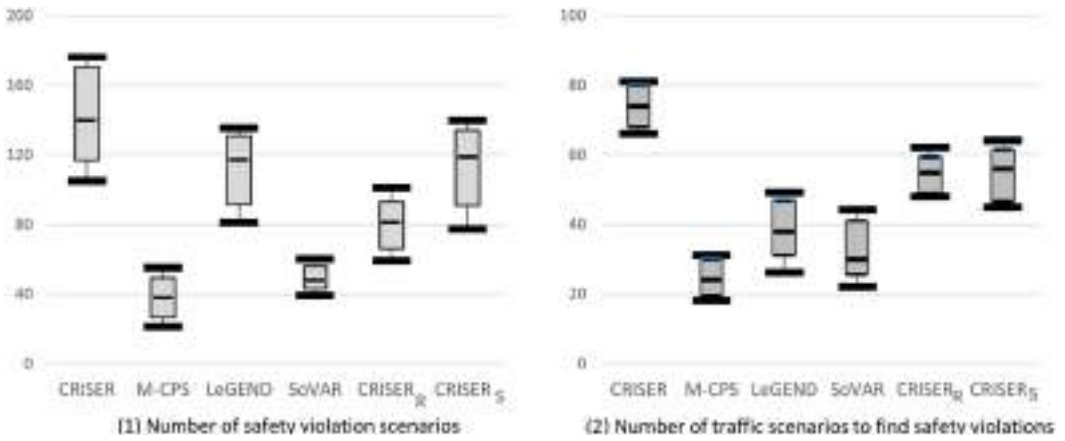


Fig. 16. The running results of CRISER and baselines.

135) safety violations of Apollo. SoVAR utilizes 33 (min 22 and max 44) accident-free scenarios to discover 49 (min 39 and max 60) safety violations of Apollo. The results demonstrate that CRISER is more effective in discovering safety violations from accident-free scenario programs than the SOTA approaches. $CRISER_R$ utilizes 54 (min 48 and max 62) accident-free scenarios to discover 80 (min 59 and max 101) safety violations of Apollo. The comparison results between CRISER and $CRISER_R$ demonstrate the effectiveness of CRISER's dual-layer exploration in generating more SVs. $CRISER_S$ utilizes 54 (min 45 and max 64) accident-free scenarios to discover 114 (min 77 and max 140) safety violations of Apollo. The comparison results between CRISER and $CRISER_S$ demonstrate the effectiveness of CRISER's semantic equivalence maintaining in discovering more types of safety violations from more accident-free scenario programs.

These experimental results demonstrate that CRISER can discover more types of safety violations in the ADS from test scenarios with initial conditions and dynamic participants' behaviors that are semantic equivalent to accident-free traffic scenarios. Compared to LeGEND, CRISER can effectively prevent non-ADS-induced violations. It can be concluded that CRISER achieves a balanced advantage by discovering more types of ADS decision-making safety violations and avoiding non-ADS-induced violations. The ablation evaluation of CRISER and $CRISER_R$ demonstrates the effectiveness of the dual-layer exploration in CRISER, and the ablation evaluation of CRISER and $CRISER_S$ demonstrates the effectiveness of the semantic equivalence maintaining in CRISER.

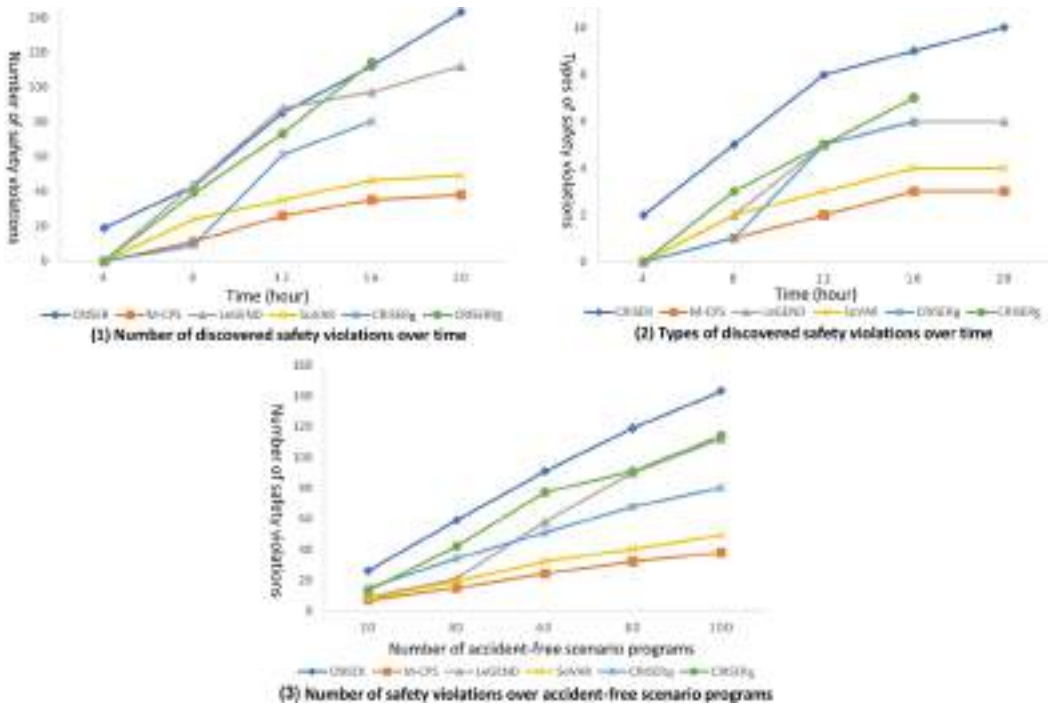


Fig. 17. The efficiency comparison results of CRISER and baselines.

Figure 17 shows the efficiency comparison of CRISER and baselines. As shown in Figure 17(1) and (2), among the 1,000 test scenarios generated by CRISER from accident-free scenario programs, CRISER takes 21 hours to discover 10 distinct types of 143 safety violations of the ADS. M-CPS takes 20 hours to generate the 1,000 test scenarios and discover 3 types of 38 safety violations of the ADS. LeGEND takes 20 hours to generate 1,000 test scenarios and discover 6 types of 112 safety violations of the ADS. SoVAR takes 29 hours to generate the 1,000 test scenarios and discover 4 types of 49 safety violations of the ADS. The results demonstrate that CRISER can generate more SVs from the same accident-free scenario programs, and reveal more types of safety violations of the ADS than the three SOTA approaches over the same time period. $CRISER_R$ takes 16 hours to generate 1,000 test scenarios, and discovers 6 types of 80 safety violations of the ADS. The comparison results between CRISER and its ablation variants demonstrate the efficiency of CRISER's dual-layer exploration and semantic equivalence maintaining in discovering more types of safety violations of the ADS from accident-free scenario programs. Figure 17(3) shows the growth in the number of SVs discovered by CRISER over the number of traffic scenarios. It can be seen that CRISER demonstrates a stable capability to generate SVs of ADS's decision-making from accident-free traffic scenarios.

6 Threats to Validity

Dataset Selection. One primary threat to validity is the selection of the real-traffic video dataset. We select Honda Scenes [39], a large-scale dataset that contains 80 hours of diverse high-quality driving video data clips. They are collected by vehicle cameras, encompassing various types of roads and participant behaviors. With the growing popularity of vehicle cameras, the videos generated directly from vehicle cameras are considered one of the most reliable methods for traffic data collection [6]. We believe CRISER can be readily applied to other datasets of traffic video recordings.

Count for Randomness. To count for its randomness, we conducted the experiments five times, and the results of each run exhibited slight variations. The running time of each experiment is long enough, and there is little difference among the experimental results of five runs. We provide the statistics results for comparison. Therefore, the experimental results can demonstrate the ability of CRISER in comparison to baselines.

7 Discussion

7.1 Limitation of Testing Assertions

The testing assertions in this work are restricted to collision, safe distance-keeping, and goal reachability. The restriction is mainly due to the methodological focus, the nature of the human-driving data we leverage, and the extensibility of our methodology design.

- Methodological focus: Our work aims to discover decision-making safety violations that fundamentally reveal decision-making failures in high-level reasoning and planning. Among various possible assertions, collision, safe distance-keeping, and goal reachability directly indicate whether the ADS can maintain safety margins and complete driving objectives without crashes.
- Human-driving scenarios as the source of initial data: CRISER uses accident-free human-driving scenarios as the initial seeds for exploration. They only guarantee that human drivers reach their goals safely without accidents, but do not ensure compliance with fine-grained traffic regulations. Consequently, CRISER is naturally aligned to explore deviations that lead to decision-level safety violations rather than rule-based behavioral deviations. Extending to rule-based assertions would require rule-annotated dataset.
- Extensibility toward other assertions: The current assertion set does not limit the generality of our approach. Other assertions can be supported by supplying human-driving scenarios that involve those specific traffic regulations and customizing the search function and fitness objectives in the outer-layer exploration process.

7.2 Limitation of Discovered SVs

In this work, the original dataset used by CRISER exclusively consist of accident-free human-driving scenarios. In these common traffic scenarios, pedestrians typically cross the road only when surrounding vehicles stop or maintain an enough safe distance. Consequently, the motions of pedestrians are inherently safe for human drivers, and no safety-critical interactions are present during pedestrians' motion process. During the scenario generation and exploration process, CRISER preserves the semantic equivalence of the original scenarios, thus the pedestrians do not impose decision-making challenges to the ADS. The framework of CRISER can be applied to traffic datasets containing scenarios with interactions involving pedestrians (e.g., the camera vehicle drives through a crowd or encounters pedestrians crossing), then it would evolve safety-critical scenarios to discover ADS safety violations involving crossing pedestrians.

8 Conclusion

In this article, we propose a novel approach, CRISER, that extracts scenario semantics and generates test scenarios from accident-free traffic videos, and discovers safety violations of the decision-making of ADS in test scenarios semantically equivalent to traffic scenarios where human-driving works safely. CRISER leverages motion change-based key frame extraction and multi-modal few-shot CoT to enhance the LMM's capability in traffic video understanding and scenario generation. Based on the extracted scenario semantics and generated test scenarios, CRISER designs a specification-based search to explore the behavior differences between ego vehicle and human-driving in semantic

equivalent scenarios. The experimental results show that CRISER can accurately extract scenario semantics and generate test scenarios from traffic videos, and effectively discover safety violations of the ADS in accident-free semantic equivalent scenarios.

References

- [1] NVIDIA. 2024. Automatically Generating Simulation Accident Scenarios for Safe and Scalable Autonomous Vehicle Testing. Retrieved July 23, 2024 from <http://nvidia.zhidx.com/content-6-3026.html>
- [2] The Autware Foundation. 2023. Autware Self-Driving Vehicle on a Highway. Retrieved September 1, 2023 from <https://www.youtube.com/watch?v=npQMzH3jd8>
- [3] AVUnit. 2024. AVUnit's documentation. Retrieved April 12, 2024 from https://avunit.readthedocs.io/en/latest/Introduction_to_AVUnit.html
- [4] GlobeNewswire. 2024. Baidu Launches Public Robotaxi Trial Operation. Retrieved April 1, 2024 from <https://www.globenewswire.com/news-release/2019/09/26/1921380/0/en/Baidu-Launches-Public-Robotaxi-Trial-Operation.html>
- [5] Business Wire. 2024. Navigent Research Names Waymo, Ford Autonomous Vehicles, Cruise, and Baidu the Leading Developers of Automated Driving Systems. Retrieved April 1, 2024 from <https://www.businesswire.com/news/home/20200407005119/en/Navigent-Research-Names-Waymo-Ford-Autonomous-Vehicles>
- [6] Miovision. 2024. Trusted Data for Mobility Planning-Portable Data Collection. Retrieved July 28, 2024 from <https://miovision.com/solutions/data-collection-traffic-studies/>
- [7] Waymo. 2024. WAYMO's Virtual World to Test Self-Driving Cars: Simulation City. Retrieved July 23, 2024 from <https://www.d1ev.com/news/jishu/150890>
- [8] GitHub. 2013. An Open Autonomous Driving Platform. Retrieved March 16, 2022 from <https://github.com/ApolloAuto/apollo>
- [9] NHTSA. 2022. Description of Light-Vehicle Pre-Crash Scenarios for Safety Applications Based On Vehicle-to-Vehicle Communications. Retrieved May 11, 2022 from <https://www.nhtsa.gov/sites/nhtsa.gov/files/811731.pdf>
- [10] GitHub. 2023. SORA-SVL Simulator. Retrieved July 30, 2024 from <https://github.com/YuqiHuai/SORA-SVL>
- [11] Weixin. 2024. Five Consecutive Tests of GPT-4V's Recognition in Autonomous Driving Scenarios. Retrieved July 28, 2024 from https://mp.weixin.qq.com/s?__biz=MzIzNjc1NzUzMw==&mid=2247699188&idx=1&sn=e4b7957166950a52a4be69cd809cf1dd&scene=21#wechat_redirect
- [12] Hugging Face. 2024. Gemma-3-1b-pt. Retrieved March 10, 2025 from <https://huggingface.co/google/gemma-3-1b-pt>
- [13] OpenAI. 2024. GPT-4V(ision) System Card. Retrieved July 28, 2024 from https://cdn.openai.com/papers/GPTV_System_Card.pdf
- [14] OpenAI. 2024. GPT-4V(ision) Technical Work and Authors. Retrieved July 28, 2024 from <https://cdn.openai.com/contributions/gpt-4v.pdf>
- [15] Weixin. 2024. GPT-4V's Answer for Autonomous Driving Corner Case Recognition. Retrieved July 28, 2024 from <https://mp.weixin.qq.com/s/IV1BXmRCFwQs2CNXDknA8Q>
- [16] Hugging Face. 2024. Llama-3.2-11B-Vision. Retrieved March 10, 2025 from <https://huggingface.co/meta-llama/Llama-3.2-11B-Vision>
- [17] GitHub. 2024. The Source Code of CRISER. Retrieved November 30, 2025 <https://github.com/ADStesting-test/CRISER>
- [18] GitHub. 2025. Autware Universe. Retrieved October 20, 2025 from https://github.com/autwarefoundation/autware_universe
- [19] Deema Almaskati, Sharareh Kermanshachi, and Apurva Pamidimukkula. 2023. Autonomous vehicles and traffic accidents. *Transportation Research Procedia* 73 (2023), 321–328.
- [20] James M. Anderson, Kalra Nidhi, Karlyn D. Stanley, Paul Sorensen, Constantine Samaras, and Oluwatobi A. Oluwatola. 2014. *Autonomous Vehicle Technology: A Guide for Policymakers*. Rand Corporation.
- [21] Earl T. Barr, Mark Harman, Phil McMinn, Muzammil Shahbaz, and Shin Yoo. 2014. The oracle problem in software testing: A survey. *IEEE Transactions on Software Engineering* 41, 5 (2014), 507–525. DOI: <https://doi.org/10.1109/TSE.2014.2372785>
- [22] Herbert Bay, Andreas Ess, Tinne Tuytelaars, and Luc Van Gool. 2008. Speeded-up robust features (SURF). *Computer Vision and Image Understanding* 110, 3 (2008), 346–359.
- [23] Yao Deng, Zhi Tu, Jiaohong Yao, Mengshi Zhang, Tianyi Zhang, and Xi Zheng. 2025. TARGET: Traffic rule-based test generation for autonomous driving via validated LLM-guided knowledge extraction. *IEEE Transactions on Software Engineering* 51, 7 (2025), 1950–1968.
- [24] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. 2017. CARLA: An open urban driving simulator. In *Proceedings of the Conference on Robot Learning*, 1–16.
- [25] Naveed Ejaz, Irfan Mehmood, and Sung Wook Baik. 2013. Efficient visual attention based framework for extracting key frames from videos. *Signal Processing: Image Communication* 28, 1, (2013), 34–44.

- [26] Y. Feng, Z. Xia, A. Guo, and Z. Chen. 2021. Survey of testing techniques of autonomous driving software. *Journal of Image and Graphics* 26, 1 (2021), 13–27.
- [27] Daniel J. Fremont, Tommaso Dreossi, Shromona Ghosh, Xiangyu Yue, Alberto L. Sangiovanni-Vincentelli, and Sanjit A. Seshia. 2019. Scenic: A language for scenario specification and scene generation. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation*, 63–78.
- [28] Lex Fridman. 2018. Human-centered autonomous vehicle systems: Principles of effective shared autonomy. arXiv:1810.01835. Retrieved from <https://arxiv.org/abs/1810.01835>
- [29] Alessio Gambi, Tri Huynh, and Gordon Fraser. 2019. Generating effective test cases for Self-Driving cars from police reports. In *Proceedings of the ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*.
- [30] Hadi Ghahremannezhad, Chengjun Liu, and Hang Shi. 2022. Traffic surveillance video analytics: A concise survey. In *Proceedings of the 18th International Conference on Machine Learning and Data Mining (MLDM '22)*, 263–291.
- [31] Sorin Grigorescu, Bogdan Trasnea, Tiberiu Cocias, and Gigel Macesanu. 2020. A survey of deep learning techniques for autonomous driving. *Journal of Field Robotics* 37, 3 (2020), 362–386.
- [32] An Guo, Yuan Zhou, Haoxiang Tian, Chunrong Fang, Yunjian Sun, Weisong Sun, Xinyu Gao, Anh Tuan Luu, Yang Liu, and Zhenyu Chen. 2024. SoVAR: Building generalizable scenarios from accident reports for autonomous driving testing. arXiv:2409.08081. Retrieved from <https://arxiv.org/abs/2409.08081>
- [33] Hardi Hungar. 2018. Scenario-based validation of automated driving systems. In *Proceedings of the International Symposium on Leveraging Applications of Formal Methods*. Springer, 449–460.
- [34] Noriah Mohd Ishak and Abu Yazid Abu Bakar. 2014. Developing sampling frame for case study: Challenges and conditions. *World Journal of Education* 4, 3 (2014), 29–35.
- [35] Jean-Michel Jullien, Christian Martel, Laurence Vignollet, and Maia Wentland. 2009. OpenScenario: A flexible integrated environment to develop educational activities based on pedagogical scenarios. In *Proceedings of the 2009 9th IEEE International Conference on Advanced Learning Technologies*. IEEE, 509–513.
- [36] Guanpeng Li, Yiran Li, Saurabh Jha, Timothy Tsai, Michael Sullivan, Siva Kumar Sastry Hari, Zbigniew Kalbarczyk, and Ravishankar Iyer. 2020. AV-FUZZER: Finding safety violations in autonomous driving systems. In *Proceedings of IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 25–36. DOI: <https://doi.org/10.1109/ISSRE5003.2020.00012>.
- [37] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the International Conference on Machine Learning*. PMLR, 12888–12900.
- [38] Wei Li, Hehe Fan, Yongkang Wong, Mohan S. Kankanhalli, and Yi Yang. 2024. TOPA: Extending large language models for video understanding via text-only pre-alignment. In *Advances in Neural Information Processing Systems*, Vol. 37 (2024), 5697–5738.
- [39] Athma Narayanan, Isht Dwivedi, and Behzad Dariush. 2019. Dynamic traffic scene classification with space-time coherence. arXiv:1905.12708. Retrieved from <https://arxiv.org/abs/1905.12708>
- [40] Hendrik Neumann and Barbara Deml. 2011. The two-point visual control model of steering-new empirical evidence. In *Proceedings of the 3rd International Conference on Digital Human Modeling (ICDHM '11)*. Springer, 493–502.
- [41] Aurélien Plyer, Guy Le Besnerais, and Frédéric Champagnat. 2016. Massively parallel Lucas Kanade optical flow for real-time video processing applications. *Journal of Real-Time Image Processing* 11 (2016), 713–730.
- [42] Vasili Ramanishka, Yi-Ting Chen, Teruhisa Misu, and Kate Saenko. 2018. Toward driving scene understanding: A dataset for learning driver behavior and causal reasoning. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [43] Esteban Rivera, Jannik Lübberstedt, Nico Uhlemann, and Markus Lienkamp. 2025. Scenario understanding of traffic scenes through large visual language models. In *Proceedings of the Winter Conference on Applications of Computer Vision*, 1037–1045.
- [44] Jonathan Roberts, Kai Han, Neil Houlsby, and Samuel Albanie. 2024. SciFIBench: Benchmarking large multimodal models for scientific figure interpretation. arXiv:2405.08807. Retrieved from <https://arxiv.org/abs/2405.08807>
- [45] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, et al. 2024. Mathematical discoveries from program search with large language models. *Nature* 625, 7995 (2024), 468–475.
- [46] Dario D. Salvucci and Rob Gray. 2004. A two-point visual control model of steering. *Perception* 33, 10 (2004), 1233–1248.
- [47] Anik Sarker, Anirban Sinha, and Nilanjan Chakraborty. 2020. On screw linear interpolation for point-to-point path planning. In *Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 9480–9487.
- [48] Huang-Chia Shih. 2013. A novel attention-based key-frame determination method. *IEEE Transactions on Broadcasting* 59, 3 (2013), 556–562.
- [49] Santokh Singh. 2015. *Critical Reasons for Crashes Investigated in the National Motor Vehicle Crash Causation Survey*.

- Traffic Safety Facts Crash Stats Report No. DOT HS 812 506. National Highway Traffic Safety Administration.
- [50] Shiming Song, Pengjun Wang, Ali Asghar Heidari, Mingjing Wang, Xuehua Zhao, Huiling Chen, Wenming He, and Suling Xu. 2021. Dimension decided Harris Hawks optimization with Gaussian mutation: Balance analysis and diversity patterns. *Knowledge-Based Systems* 215 (2021), 106425.
 - [51] Shuncheng Tang, Zhenya Zhang, Jixiang Zhou, Lei Lei, Yuan Zhou, and Yinxing Xue. 2024. LeGEND: A top-down approach to scenario generation of autonomous driving systems assisted by large language models. arXiv:2409.10066. Retrieved from <https://arxiv.org/abs/2409.10066>
 - [52] Haoxiang Tian, Yan Jiang, Guoquan Wu, Jiren Yan, Jun Wei, Wei Chen, Shuo Li, and Dan Ye. 2022. MOSAT: Finding safety violations of autonomous driving systems using multi-objective genetic algorithm. In *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 94–106.
 - [53] Haoxiang Tian, Guoquan Wu, Jiren Yan, Yan Jiang, Jun Wei, Wei Chen, Shuo Li, and Dan Ye. 2022. Generating critical test scenarios for autonomous driving systems via influential behavior patterns. In *Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering*, 1–12.
 - [54] Tian Wang, Meina Qiao, Aichun Zhu, Guangcun Shan, and Hichem Snoussi. 2020. Abnormal event detection via the analysis of multi-frame optical flow information. *Frontiers of Computer Science* 14 (2020), 304–313.
 - [55] Yi Wang, Xinhao Li, Ziang Yan, Yinan He, Jiashuo Yu, Xiangyu Zeng, Chenting Wang, Changlian Ma, Haian Huang, Jianfei Gao, et al. 2025. InternVideo2. 5: Empowering video MLLMs with long and rich context modeling. arXiv:2501.12386. Retrieved from <https://arxiv.org/abs/2501.12386>
 - [56] Xudong Zhang and Yan Cai. 2023. Building critical testing scenarios for autonomous driving from real accidents. In *Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis*, 462–474.
 - [57] Maximilian Zipfl, Nina Koch, and J. Marius Zöllner. 2023. A comprehensive review on ontologies for scenario-based testing in the context of autonomous driving. In *Proceedings of the 2023 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 1–7.

Appendices

A Specifications for Initial Conditions and Behaviors of Dynamic Participants

A.1 Specifications for Participant Behaviors

Types of participant behaviors include: follow lane, change left/right, turn left/right, cross, accelerate, decelerate, brake, walk along, walk across, and stand. Their specifications are given as follows.

A.1.1 Specification for Follow Lane.

- (1) *Lane specification*: $(pos^0, pos^F \in l_s) \wedge (l_s \in R) \wedge (fd(pos^0, pos^F) = df(l_s))$, where pos^0 is the position of initial waypoint on the trajectory and pos^F is the last waypoint. l_s is a lane on the road R . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on the same lane l_s , and the direction from starting position to ending position is along l_s .
- (2) *Waypoint specification*: $\forall i \in (0, F) \ pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)]$, $pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of the waypoint in the trajectory segment. This specification verifies that the positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) = 0$. This specification verifies that the driving directions of intermediate waypoints are along the lane.
- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.2 Specification for Change Left.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, (s \neq f) \wedge (l_s, l_f \in R), (df(l_s) = df(l_f)) \wedge (-1 < \sin(fd(pos^0, pos^F), df(l_s)) < 0)$, where pos^0 is the initial waypoint of the trajectory and

pos^F is the last waypoint. l_s and l_f are two lanes on the road R . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on two different lanes on the same road, and the directions of two lanes are the same. The ending position is on the left lane of starting position.

- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (90, 180)$. This specification verifies that the driving directions are to the left.
- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.3 Specification for Change Right.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, (s \neq f) \wedge (l_s, l_f \in R), (df(l_s) = df(l_f)) \wedge (0 < \sin(fd(pos^0, pos^F), df(l_s)) < 1)$, where pos^0 is the initial waypoint on the trajectory and pos^F is the last waypoint. l_s and l_f are two lanes on the road R . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on two different lanes on the same road, and the directions of two lanes are the same. The ending position is on the right lane of starting position.
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (-90, 0)$. This specification verifies that the driving directions are to the right.
- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.4 Specification for Turn Left.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, l_s \in R_1, l_f \in R_2, (R_1 \text{ not } = R_2) \wedge (-1 < \sin(df(l_s), df(l_f)) < 0) \wedge (-1 < \sin(fd(pos^0, pos^F), df(l_s)) < 0)$, where pos^0 is the position of the initial waypoint of the trajectory and pos^F is the last waypoint. l_s and l_f are two lanes on different roads R_1 and R_2 . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on different roads, and the direction of ending lane is on the left of the direction of starting lane.
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (90, 180)$. This specification verifies that the driving directions are to the left.

- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.5 Specification for Turn Right.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, l_s \in R_1, l_f \in R_2, (R_1 \text{ not } = R_2) \wedge (0 < \sin(df(l_s), df(l_f)) < 1) \wedge (0 < \sin(fd(pos^0, pos^F), df(l_s)) < 1)$, where pos^0 is the initial waypoint of the trajectory and pos^F is the last waypoint. l_s and l_f are two lanes on different roads R_1 and R_2 . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on the lanes of different roads, and the direction of ending lane is on the right of the direction of starting lane.
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (-90, 0)$. This specification verifies that the driving directions are to the right.
- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.6 Specification for Cross.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, l_s \in R_1, l_f \in R_2, (R_1 \text{ not } = R_2) \wedge (\sin(df(l_s), df(l_f)) = 0)$, where pos^0 is the position of the initial waypoint on the trajectory and pos^F is the last waypoint. l_s and l_f are two lanes on different roads R_1 and R_2 . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on the lanes of different roads, and the direction of ending lane is the same as the direction of starting line.
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (0, 180)$. This specification verifies that the driving directions are not opposite to the direction of starting and ending lanes.
- (4) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t < threshold_c$, where $speed_{max}$ is the speed limits of the road, and $threshold_c$ is the threshold for speed change during the actions except accelerating, decelerating and braking.

A.1.7 Specification for Accelerate.

- (1) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (0, 180)$. This specification verifies that the driving directions are not opposite to the direction of starting and ending lanes.
- (2) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|)/\Delta t \geq threshold_{ac}$, where $speed_{max}$ is the speed limits of the road, and $threshold_{ac}$ is the minimal threshold for speed change during accelerating.

A.1.8 Specification for Decelerate.

- (1) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (0, 180)$. This specification verifies that the driving directions are not opposite to the direction of starting and ending lanes.
- (2) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|) / \Delta t \leq -1 * threshold_{ac}$, where $speed_{max}$ is the speed limits of the road, and $-1 * threshold_{ac}$ is the maximal threshold for speed change during decelerating, which is a negative numeric value.

A.1.9 Specification for Brake.

- (1) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (0, 180)$. This specification verifies that the driving directions are not opposite to the direction of starting and ending lanes.
- (2) *Speed specifications*: $\forall i \in (0, F), (|vel^i| \leq speed_{max}) \wedge (|vel^{i+1}| - |vel^i|) / \Delta t \leq -1 * threshold_{ac} \wedge |vel^F| = 0$, where $speed_{max}$ is the speed limits of the road, and $-1 * threshold_{ac}$ is the maximal threshold for speed change during decelerating, which is a negative numeric value. Based on the decelerate identification criteria, if the speed at the time when the action ends is zero, the action is identified as brake.

A.1.10 Specification for Walk along.

- (1) *Lane specification*: $(pos^0, pos^F \in l_s) \wedge (l_s \in R) \wedge (fd(pos^0, pos^F) = df(l_s))$, where pos^0 is the position of initial waypoint on the trajectory and pos^F is the last waypoint. l_s is a lane on the road R . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on the same lane l_s , and the direction from starting position to ending position is along l_s .
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of the waypoint in the trajectory segment. This specification verifies that the positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) = 0$. This specification verifies that the walking directions of intermediate waypoints are along the lane.
- (4) *Speed specifications*: $\forall i \in (0, F), |vel^i| \leq peds_{max}$, where $peds_{max}$ is the maximal speed.

A.1.11 Specification for Walk Across.

- (1) *Lane specification*: $pos^0 \in l_s, pos^F \in l_f, l_s \in R_1, l_f \in R_2, (R_1 \text{ not } = R_2) \wedge (\sin(df(l_s), df(l_f))) = 0$, where pos^0 is the position of the initial waypoint on the trajectory and pos^F is the last waypoint. l_s and l_f are two lanes on different roads R_1 and R_2 . df represents the direction and $fd(A, B)$ is the angle from direction A to direction B. This specification verifies that the starting position and ending position are on the lanes of different roads, and the direction of ending lane is the same as the direction of starting line.
- (2) *Waypoint specification*: $\forall i \in (0, F), pos^i \in l_s \cup l_f, pos^i.x \in [\min(pos^0.x, pos^F.x), \max(pos^0.x, pos^F.x)], pos^i.y \in [\min(pos^0.y, pos^F.y), \max(pos^0.y, pos^F.y)]$, where i represents the index of waypoint in the trajectory segment. This specification verifies that positions of intermediate waypoints of the trajectory segment are between the starting position and ending position.
- (3) *Direction specification*: $\forall i \in (0, F), \arctan2(vel_y^i, vel_x^i) \in (0, 180)$. This specification verifies that the walking directions are not opposite to the direction of starting and ending lanes.
- (4) *Speed specifications*: $\forall i \in (0, F), |vel^i| \leq peds_{max}$, where $peds_{max}$ is the maximal speed.

A.1.12 Specification for Stand.

- (1) *Waypoint specification*: $\forall i \in (0, F], pos^i = pos^0$, where i represents the index of waypoint in the trajectory segment. This specification verifies that the positions of waypoints are the same throughout the action period.
- (2) *Speed specifications*: $\forall i \in (0, F), |vel^i| = 0$. This specification verifies that the speeds of waypoints are zero throughout the action period.

A.2 Specifications for Relative Position

The relative positions include: ahead, right front, left front, opposite, left vertical, and right vertical. The specifications for them are given as follows.

- (1) *Ahead*: This specification verifies the initial position of the participant's behavior is in front of the initial position I of ego vehicle on the same lane. $l'_j = l'_i, \cos(d(l'_i), fd(I, pos^0)) = 1, I \in l'_i, pos^0 \in l'_j$
- (2) *Right front*: This specification verifies the initial position of the participant's behavior is in the right front of the initial position I of ego vehicle. $0 < \sin(fd(I, pos^0), d(l'_i)) < 1, I \in l'_i$
- (3) *Left front*: This specification verifies the initial position of the participant's behavior is in the left front of the initial position I of ego vehicle. $-1 < \sin(fd(I, pos^0), d(l'_i)) < 0, I \in l'_i$
- (4) *Opposite*: This specification verifies the initial position of the participant's behavior is at the opposite side of the intersection where the initial position I of ego vehicle is. $l'_j \neq l'_i, \cos(d(l'_i), fd(I, pos^0)) = 1, I \in l'_i, pos^0 \in l'_j$
- (5) *Right vertical*: This specification verifies the initial position of the participant's behavior is in the right side of the intersection where the initial position I of ego vehicle is. $\sin(d(l'_j), d(l'_i)) = -1, 0 < \sin(fd(I, pos^0), d(l'_i)) < 1, I \in l'_i, pos^0 \in l'_j$
- (6) *Left vertical*: This specification verifies the initial position of the participant's behavior is in the left side of the intersection where the initial position I of ego vehicle is. $\sin(d(l'_j), d(l'_i)) = 1, -1 < \sin(fd(I, pos^0), d(l'_i)) < 0, I \in l'_i, pos^0 \in l'_j$

B Parameter Sensitivity Analysis for CRISER's Scenario Video Information Extraction Pipeline

RA represents the extraction accuracy of road, EA represents the extraction accuracy of ego driving task, PA represents the extraction accuracy of participants, and RPA represents the extraction accuracy of their relative positions.

Table B1. The Joint Configuration Experimental Results of the Parameter Sensitivity Analysis for CRISER's Scenario Video Information Extraction Pipeline

ID	α	β	b_f	b_r	RA	EA	PA	RPA	ID	a	b	bf	br	RA	EA	PA	RPA
1	0.5	1.0	1%	10	0.96	0.94	0.60	0.54	8	0.2	1.75	1%	10	0.96	0.94	0.76	0.68
2	0.4	1.0	1%	10	0.96	0.94	0.60	0.56	9	0.2	1.5	2%	10	0.96	0.94	0.84	0.76
3	0.3	1.0	1%	10	0.96	0.94	0.68	0.64	10	0.2	1.5	3%	10	0.96	0.94	0.80	0.72
4	0.2	1.0	1%	10	0.96	0.94	0.74	0.68	11	0.2	1.5	4%	10	0.96	0.94	0.78	0.66
5	0.1	1.0	1%	10	0.96	0.94	0.68	0.62	12	0.2	1.5	2%	20	0.96	0.94	0.88	0.78
6	0.2	1.25	1%	10	0.96	0.94	0.78	0.70	13	0.2	1.5	2%	30	0.96	0.94	0.94	0.84
7	0.2	1.5	1%	10	0.96	0.94	0.80	0.74	14	0.2	1.5	2%	40	0.96	0.94	0.90	0.80

Bold text indicates the approach's parameter configuration that achieves the best performance among the 14 rounds of parameter experiments.

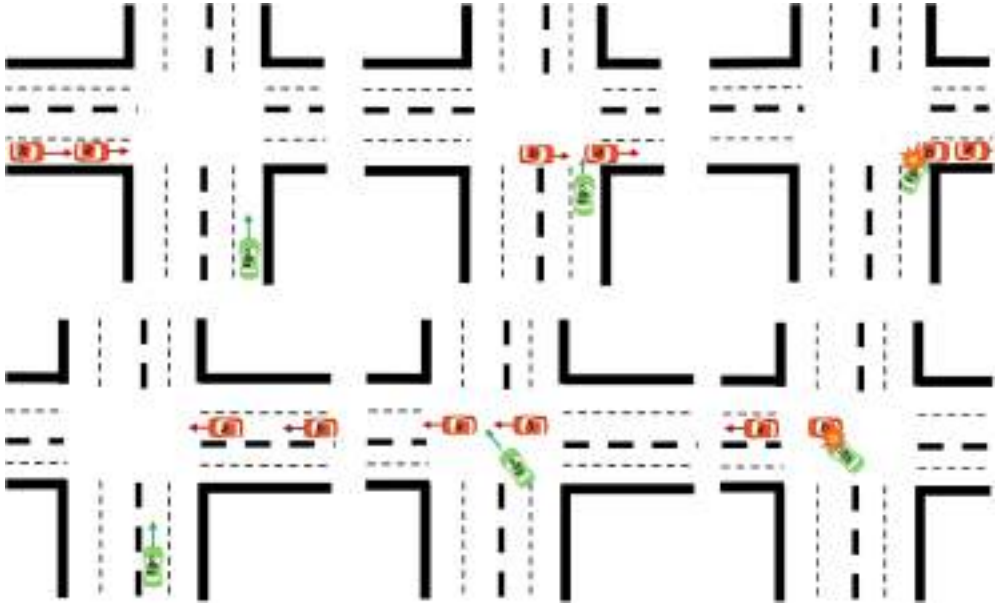


Fig. C1. Two examples of safety violation 1.

C Examples of Each Type of Discovered SV

C.1 Examples of Safety Violation 1

As shown in Figure C1, in the upper scenario, the driving task of EV and human-driving is *cross the junction*, EV is turning right. Vehicle *a* is driving from the left vertical lanes to cross the intersection, and vehicle *b* is following *a* to cross the intersection. EV misidentifies the speeds of *a* and *b* as the same, ignoring the acceleration of vehicle *b*. EV stops to let *a* pass and then accelerates to turn right, causing to collide with vehicle *b*. The same error of EV causes the safety violation of the lower scenario, the driving task of EV and human-driving is *turn left*. For the human-driving in the two traffic scenarios, it waits to let the continuous vehicles that are entering the same lane and occupy the right of way at high speeds to pass by, and then continues to enter the incoming lane.

C.2 Examples of Safety Violation 2

As shown in Figure C2, the driving task of EV and human-driving is “*change lane*”. EV is driving along the lane to go through the road. Vehicle *a* ahead of EV is also driving along the lane, and vehicle *b* is driving behind on the adjacent lane. As *a* decelerates, EV does not plan to change to the right lane to overtake vehicle *a* at first time. It hesitates the overtaking route decision for many times. When EV starts to change right, vehicle *b* accelerates to drive through. However, EV ignores the speed change of vehicle *b* from behind, causing it to collide with vehicle *b* which has the right of way. For human-driving in the traffic scenario, it decelerates and starts to change to the right lane for overtaking at first time when vehicle *a* significantly decelerates. When vehicle *b* accelerates. Thus it further decelerates to let vehicle *b* from behind on the adjacent lane pass through.

C.3 Examples of Safety Violation 3

As shown in Figure C3, the driving task of EV and human-driving is “*cross the intersection*.” EV is crossing the intersection. The vehicle *a* and vehicle *b* are driving on the left vertical lanes to cross

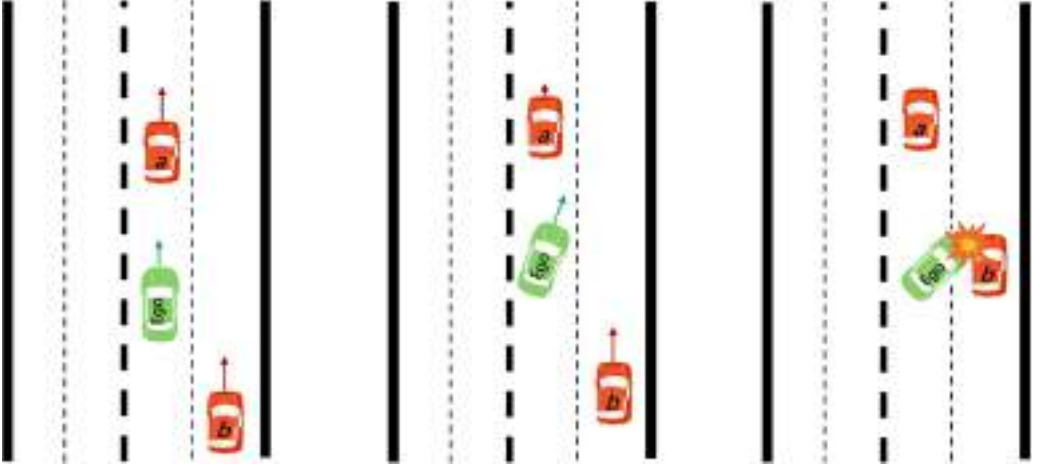


Fig. C2. An example of safety violation 2.

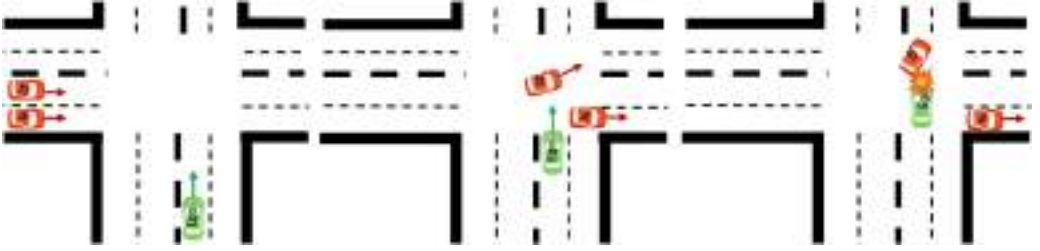


Fig. C3. An example of safety violation 3.

the intersection. When EV enters the connection area of the junction, it predicts the behavior of *a* correctly and stops to let *a* pass. However, EV mispredicts the behavior of vehicle *b* due to ignoring the behavioral changes (the turning of *b* during EV's crossing) of *b*, causing collide with vehicle *b* in the connection area while crossing the intersection. *For the human-driving in the traffic scenario*, it decelerates slightly while crossing the intersection to observe changes in other vehicles and pedestrians from other directions at the intersection.

C.4 Examples of Safety Violation 4

As shown in Figure C4, the driving task of EV and human-driving is "turn left." EV is turning left at the intersection, and the vehicle *a* is driving on the left vertical lanes. While EV turning left in the connection area of the intersection, vehicle *a* is turning left. The distributions of the lanes in the connection area are complicated, facing with them, EV fails to handle the right-of-way of the vehicle on the lanes in this area, leading to colliding with vehicle *a*. For the human-driving in the traffic scenario, it decelerates during turning to let vehicle *a* pass and then continues the turning.

C.5 Examples of Safety Violation 5

As shown in Figure C5, the driving task of EV and human driver is driving along the lane to go through. The vehicle *a* is driving slowly in front of EV on the lane. Another vehicle *b* is driving in front of EV on the adjacent lane. In the upper scenario of Figure C5, when vehicle *a* stops, EV plans an overtaking trajectory to go around vehicle *a*. When EV changes to the lane of vehicle *b*,

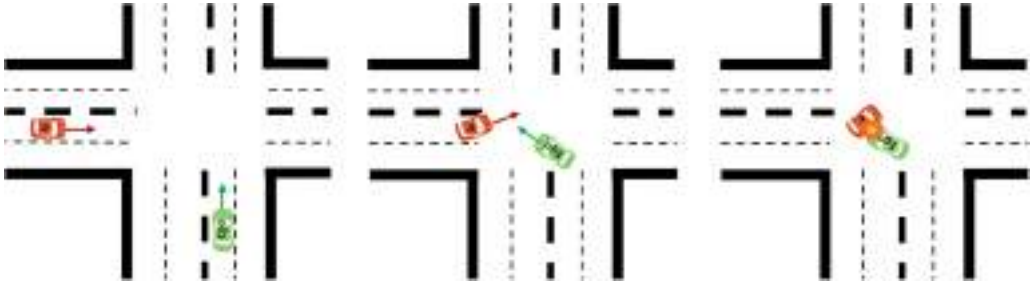


Fig. C4. An example of safety violation 4.

the vehicle *b* decelerates and stops in the front of EV. The distance between vehicle *b* and EV is enough for EV to change to the left lane to continue the driving task. However, EV fails to adjust a new trajectory to change lane again. In the lower scenario of Figure C5, when vehicle *b* is driving away for a long distance from EV, and no other NPC vehicles drive on the right lane. But EV still follows vehicle *a* with a slow speed. Vehicle *a* decelerates and stops on the lane due to vehicle malfunctions, EV adjusts its route to plan an overtaking trajectory, and starts to change to the right lane. But during changing lane, EV drives forward for a not short distance, leading to collision with vehicle *a*. For the human-driving in this traffic scenario, it starts to change to the right lane at the moment of vehicle *b* driving away, and drives forward for a short distance during changing lane. When vehicle *b* stops, it changes lane again to the original lane.

C.6 Examples of Safety Violation 6

As shown in Figure C6, the driving task of EV and human-driving is “turn right.” EV is turning right at the intersection. The vehicle *a* is driving from the left vertical lane to cross the intersection, and then brakes to stop at the incoming lane of EV. EV detects the movement of vehicle *a* but fails to make correct response to adjust its trajectory to enter another lane to complete the driving task (the adjacent lane of the incoming lane of EV also allows the turning right vehicle). For the human-driving in the traffic scenario, when the incoming lane of it is stuck by a vehicle, it changes to the adjacent lane of the originally planned incoming lane and continues to drive instead of stopping at the entrance of the lane, leading to disrupting the normal traffic flow.

C.7 Examples of Safety Violation 7

As shown in Figure C7, in the left scenario, the driving task of human driver and EV is turn left. Vehicle *a* is crossing the intersection from the opposite side lanes, and vehicle *b* is following *a* to cross the intersection. In the real-traffic video, the human driver decelerates to wait for vehicle *b* to pass by, and then accelerates to finish the turn left. In the test scenario of Apollo, before EV starts to turn at the entrance of junction, Apollo identifies the right of way of *a* and stops to let *a* pass. When vehicle *a* passes the junction, EV continues to turn left due to wrongly calculation of the distance and speed of *b*, causing collision to *b*. The safety violation of the right scenario is caused by the same error of EV. The driving task of human driver and EV is cross. In the two traffic scenarios, the human driver either crosses the connection area with acceleration to turn left before vehicle *a* approaching, or waits for vehicle *b* to pass by before turning left. EV waits for vehicle *a* to pass by but ignoring vehicle *b* approaching, leading to the collision.

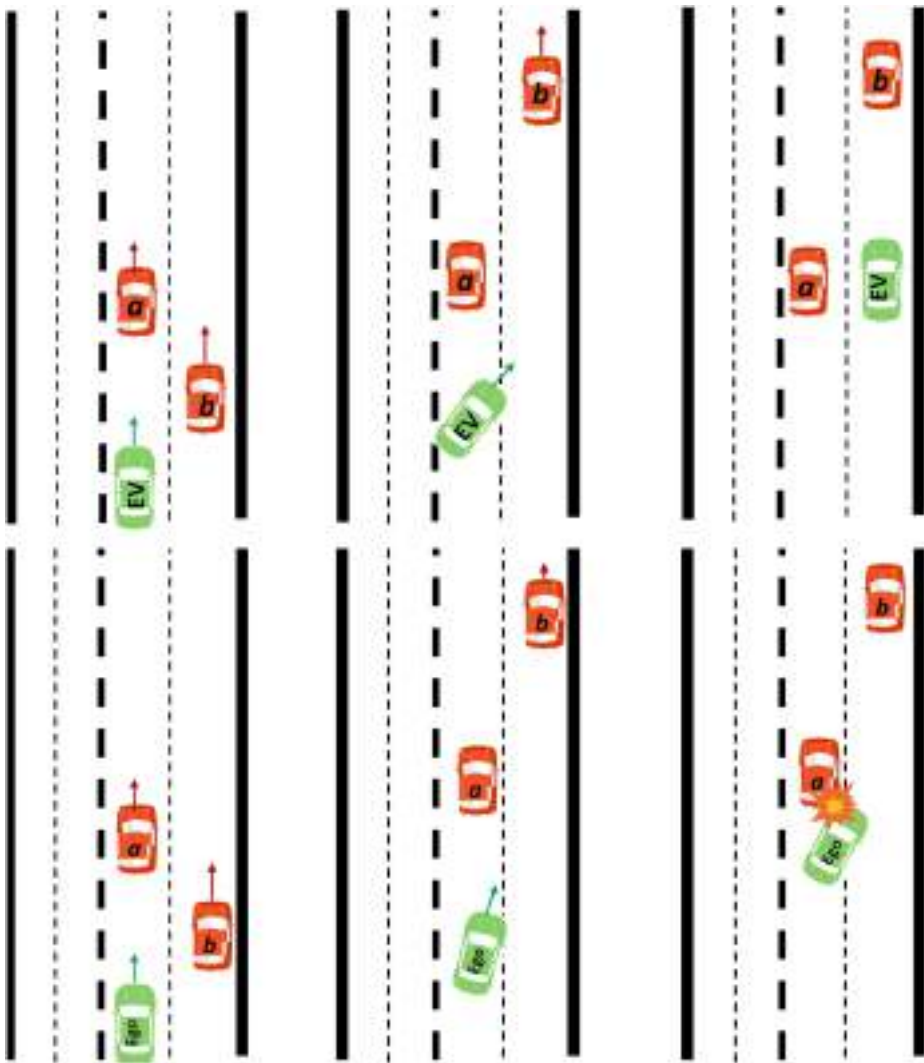


Fig. C5. An example of safety violation 5.



Fig. C6. An example of safety violation 6.

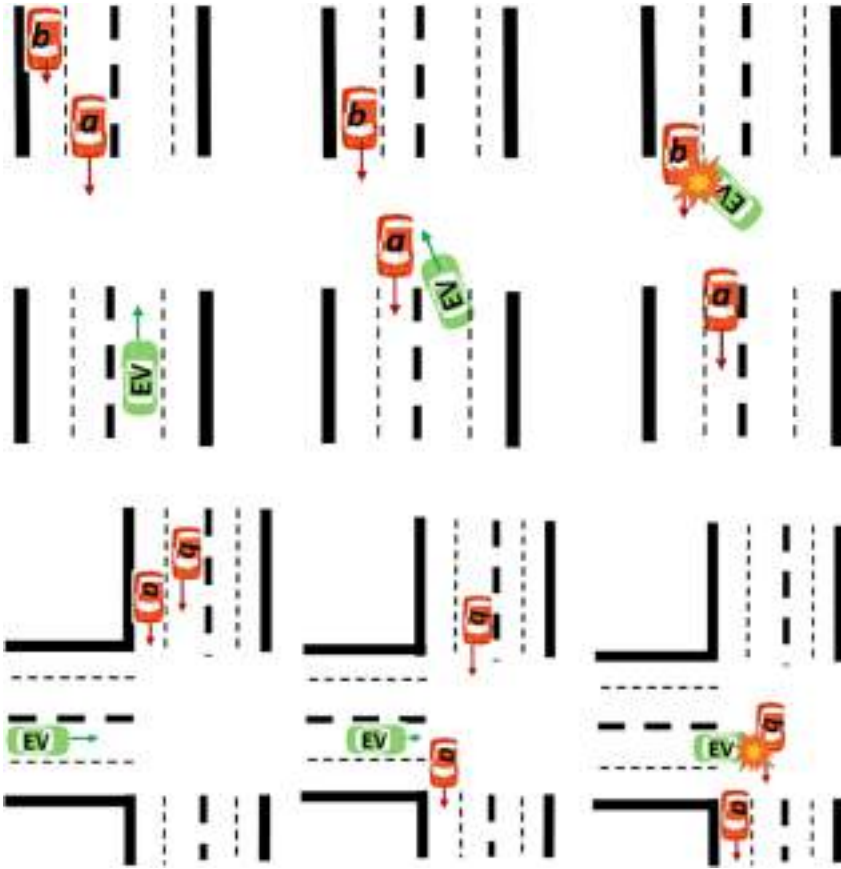


Fig. C7. Two examples of safety violation 7.

C.8 Examples of Safety Violation 8

As shown in Figure C8, the driving task of EV and human-driving is “drive through.” EV is driving behind two vehicles along the lane. Vehicle *a* and *b* are driving slowly, causing EV to drive at a low speed. After EV gets stuck in traffic for some moment, it moves too close to them. EV fails to plan a route to change lane to go around the two vehicles. For the human-driving in the traffic scenario, when the two trucks ahead on the lane block and no nearby vehicles drive on the adjacent lane, it changes to the adjacent lane and accelerates to drive away from blocked trucks.

C.9 Examples of Safety Violation 9

As shown in Figure C9, the driving task of EV and human-driving is “turn right.” EV is turning right. Truck *a* on the left vertical lanes is crossing. When EV starts to turn, *a* is driving close to the adjacent lane of EV’s destination. EV decelerates and turns right close to the right corner and keeps a safe distance with *a*. However, when EV finishes turning, it accelerates close to the lane marking, not considering the big size of *a* and leading to the collision. For human-driving in this traffic scenario, it drives not close to the lane marking, and not close to the large truck.

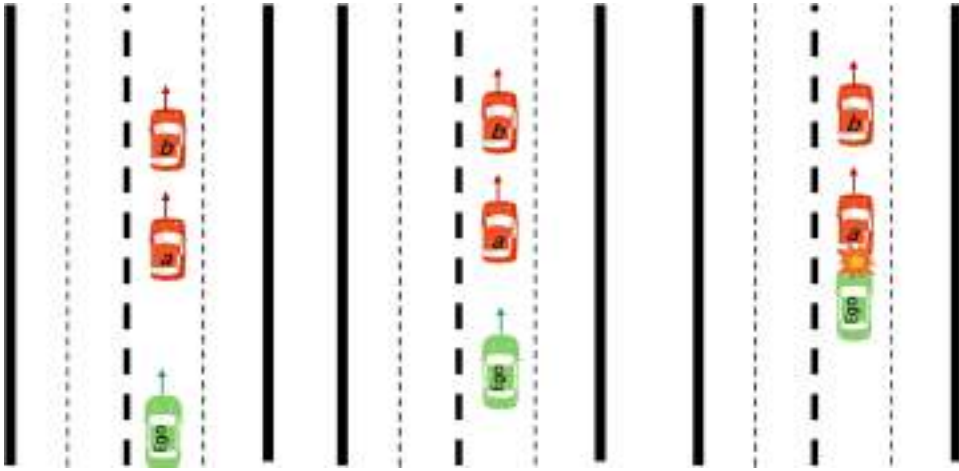


Fig. C8. An example of safety violation 8.

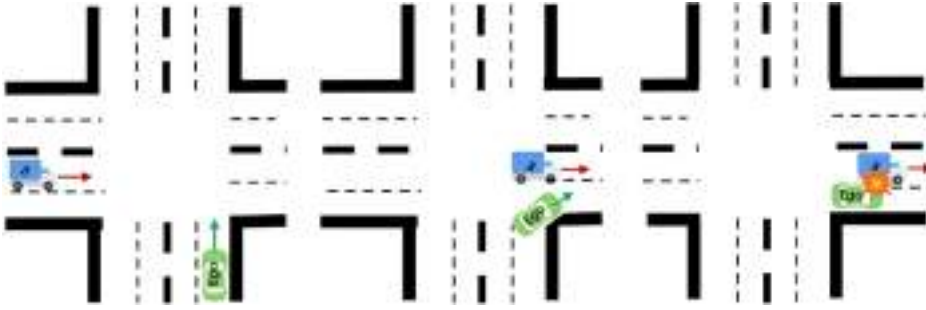


Fig. C9. An example of safety violation 9.

C.10 Examples of Safety Violation 10

As shown in Figure C10, the driving task of EV and human-driving is “cross the intersection.” EV is crossing the intersection. The vehicle *a* is driving ahead of EV and entering the next lane after the intersection. When EV is crossing the intersection, vehicle *b* on the vertical lane is keeping a long distance with EV and turning left at a slow speed, so that EV accelerates to pass by. However, when EV has crossed the intersection and starts driving on the next lane, it does not decelerate to keep distance with vehicle *a*. When vehicle *a* stops, EV rear-ends collision with it. For the human-driving in the traffic scenario, when it finishes the cross action and enters the lane of vehicle *a*, it decelerates to follow *a* or changes to the right lane to overtake *a* as the right lane is no other vehicle driving.



Fig. C10. An example of safety violation 10.

Received 11 August 2025; revised 12 November 2025; accepted 28 December 2025